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Doctoral thesis – Speciality: acoustics

Presented by:
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submitted for the Doctoral degree

Numerical modeling of metaporous layers

Computation of scattering and dispersion properties of stacks and periodic arrangements of porous layers

Defense scheduled on the ??

Members of the defense committed:

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List of Acronyms

BEM Boundary Elements Method.
FEM Finite Elements Method.
GEP Generalized Eigenvalue Problem.
JCA Johnson-Champoux-Allard.
MST Multiple Scattering Theory.
PML Perfectly Matched Layer.
SAFE Semi-analytical Finite Elements.
SCM Spectral Collocation Method.
SRR Split-Ring Resonator.

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Introduction

Noise is a major issue in our modern societies, ranging from everyday life inconveniences, work comfort problems to an actual pollution problem in severly affected areas. Sources of noise can be diverse; in urban environments, concerns are mostly on population and traffic noise. From industrial origins, noise can be of several consequences, causing not only risks to the workers and the neighbouring populations, but also structural hazard to installations. Generally, motorized transport is a major issue in all environments, including aircraft, public transportation, roads, commercial vessel traffic, etc.

Although sound intensity can be measured objectively, the impact of noise on individuals is in part shaped by social and cultural context. Studies show that expected behaviours norms and customs strongly influence reported annoyance[1], with comparative surveys revealing clear cross-cultural differences. More recent work has linked collectivistic cultural norms to systematically higher everyday noise exposure[2,3].

The European Environment Agency estimated in 2021 that 1 in 5 European resident are exposed to unhealthy levels of noise, which mostly originates from traffic noise. These noise exposures can have a strong impact on health[4] favoring impaired circadian rythm, oxidative stress and cardiovascular dysfunction[5], as well as psychological consequences like anxiety and depression[6]. Moreover, anthropic noise pollution also affects biodiversity at large, causing stress, behaviour changes, and masking sensory perception[7]. Marine traffic was shown to generate stress on marine mammals and modify their behaviour by provoking area avoidance and vocalization disruption[8].

The emergence of all these noise sources comes with the need to mitigate them up to our current capabilities. The latter is able to evolve thanks to research on noise barriers and acoustic insulation systems. Various acoustic insulation strategies have been applied throughout the ages. Dating back to 4th century BC, acoustic jars were used to improve the room acoustics in Greek and Roman theatres as well as in the Middle Ages in Western Europe in churches[9]. These resonators are thought to be capable of reducing reverberation time in buildings, covering a frequency range matching that of human spoken voice. As another example, a famous roman amphitheater was shown to have a seat row disposition able to improve intelligibility of speeches from the stage[10].

Contemporary to these antique devices, porous and fibrous materials have been traditionally used to absorb sound in the last century[11–16]. Largely used in the industry, they are efficient at frequencies starting close to the quarter wavelength resonance frequency of the layer and have been extensively analyzed in the literature[17–21]. They can be found in natural occurences or engineered and cover very large use cases of acoustic insulation such as industry, transport, aerospace, concert halls and many others. The picture of a porous material is shown in Fig. I.1a. (utile ?)

Fig. I.1: a) Image de la microstructure d'un poreux b) Coefficient de réflexion et absorption d'une couche rigid-backed d'un PEM avec un schéma. c) Coefficient de réflexion, transmission et absorption d'une couche avec transmission d'un PEM avec un schéma.

State of art

Overview of Acoustic Metamaterials

In more recent years, the emergence of more complex engineered structures has disrupted the landscape of acoustic absorbers. Metamaterials are composite structures designed to exhibit physical effective properties that go beyond the capability of the sole material that composes the bulk. They can be created by a lot of various means, including engineered microstructure, a gradial evolution of the material properties, or repeating patterns or scatterers implemented in one or several directions in space and embedded in a host material as shown for various dimensions in Fig. I.2. Compared to homogeneous layers, they generally lead to more complex structures, and are able to achieve enhanced effects. Usually allowing for a tunable response, these structures offer great flexibility and open new perspectives in material design. Therefore, they have sparked interest in various disciplines, such as photonics and optics[22]. They can be used to design antennas and absorber for electromagnetic waves, seismic protection devices among others. In acoustic, they allow for various applications such as absorption, filtering, diffusion or cloaking.

Seminal works have been done on the study of the use of lattice arrangements or crystals and have been applied to wave control application. Photonic crystals are used to achieve the control of light and electromagnetic waves[23]. Phononic crystals have been developed for the control of elastic and acoustic waves as the periodic distribution of scatterers in a solid host medium[24]. In the case where the host medium is constituted by a fluid, these structures are called sonic crystals[25]. These structures are often studied through their band structure (also called dispersion diagram when used to study non-periodic geometries). These diagram represents the wavenumber in a given direction against the frequency, thus informing on the evolution of the wave velocities in the structure and thus the dispersion. These aspects will be addressed in the subsequent chapter, especially regarding the background theory and the available numerical tools to compute them.

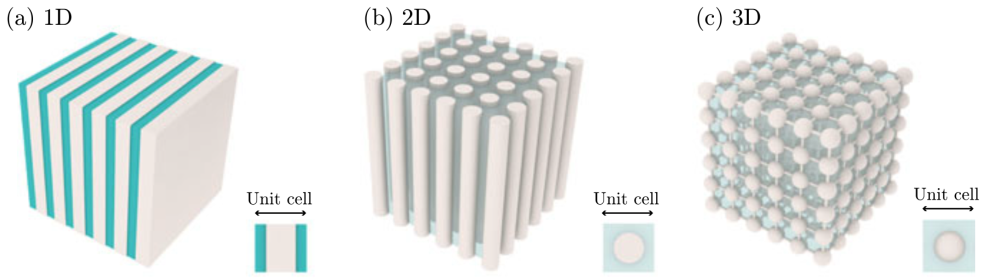

Fig. I.2: Periodic cells in 1, 2 and 3D. Image taken from Ref.[26]

In the remaining of the literature overview, focus will be centered around metamaterials that involve mostly periodicity in only one direction. These acoustic metamaterials solutions can be divided into two main categories: the first concerns the so-called *reflection problem* where the case study is that of an incoming wave on some sample or panel which has a rigid-backing at the end. There, the consideration is focused on mitigating the reflected wave and acheving important acoustic absorption. The other case is that of a *transmission problem* where the incoming wave is able to transmit through the sample. In this case and depending on the application, either or both reflection and transmission are sought to be mitigated. The symmetry of the structure is of importance in this case.

Structured materials that tackle the reflection problem can involve many various strategies. The combination of a perforated plate and coiled-up channels is able to achieve large broadband absorption underwater[27]. Micro-perforated plate can also be used to achieve interesting absorption properties for airborne acoustics[28]. Materials structured from resonant building blocks with resonance frequencies in the sub-wavelength regime i.e., for frequencies which wavelength is much bigger than the typical size of the material, have also been studied[29–31]. Large width elastic bandgaps have been achieved using locally resonant cells in the design of a sonic crystal[32]. These resonances introduce strong dispersion of the waves and the properties of the system are thus dramatically affected. Together with the unavoidable losses, the resonant behaviour of metamaterials make them perfect candidates to design efficient absorbers at low frequencies with reduced dimensions[33,34]. Balancing the dissipation and leakage of each resonance yields the recently described *critical coupling*. In this case, the reflection drops to zero[35]. Perfect absorbers can thus be designed for reflection problems for both monochromatic waves[33,36] and broad range of frequencies[36–39]. Yang *et al.* have realized the backing of a porous layer using Fabry-Pérot channels to design a very efficient broadband absorber[40].

One of the initial interests of the acoustic metamaterials community was the possibility of producing deeps in the transmission spectra at very low frequencies. This has been widely developed in the literature, a small subset of these works include Refs[41–44]. These configurations are based on resonant elements and display a zero transmission in the lossless case at their resonance frequency. In reality, losses cannot be avoided and a small amount of the energy is always transmitted. Membrane-type acoustic metamaterials[45–47], or acoustic metamaterials made by Helmholtz resonators[48–50] have been used to develop systems with high values of transmission loss for audible acoustics. Considering the vibroacoustic coupling allows for more precise depiction of real-life applications. Works on periodic soft inclusion in a soft solid matrix was done by Aberkane *et al.* showing improved transmission loss affected by the presence of soft inclusions in the solid layer. In this case, the presence of losses was able to improve the transmission loss at low frequencies. Also accounting for vibroacoustic coupling, attached resonators to walls have been also used to reduce the transmission loss of panels[51,52].

For the case of mirror-symmetric structures, it is worth noting here that if only a monopolar resonator (Helmholtz) or a dipolar resonator (membrane) is used to absorb waves, then only 50% of energy can be absorbed as shown in several articles[53,54]. However, using a material made of slits loaded by an array of identical resonators creates an accumulation point close to the resonance frequency of the resonators allowing the accumulation of the resonances of the slit, thus placing together symmetric and antisymmetric resonances[45,48,55], although requiring for a compromise between absorption and thickness. A more complex approach consists in designing degenerate resonators. By using combinations of membrane and cavity resonators[45,56] and with slits loaded with Helmholtz resonators[54], quasi-total absorption can be reached. In these systems, each resonator manages half of the problem, absorbing each one half of the energy and allowing a perfect absorption peak: all the energy that impinges on the system is dissipated within the system, i.e., the energy is neither reflected nor transmitted. The other alternative to obtain perfect absorption consists in breaking the symmetry of the system. These systems can achieve unidirectional perfect absorption, i.e., perfect absorption can be obtained from one of the two incidence sides[53]. The principle of these systems consists in using two resonators, one that is used to reduce as much as possible the transmission and the other that impedance matches the system from one side[33,36,47]. This concept has been exploited to develop broadband perfect absorbers using a cascade of the previous process giving rise to a rainbow absorber[37].

Metaporous layers

In order to increase the capabilities of acoustic absorption of porous materials, some works can involve the modification of the core structure of porous and poroelastic materials: tuning of the microstructure, gradient properties, anisotropic materials. Recently, with the increase of new fabrication techniques based on 3D printing, the structure of the porous materials can be controlled and tuned to design materials with better absorption capabilities at low frequencies, based on gradient properties[57,58], or on anisotropic materials[59–62]. Recently, resonators filled with 3D printed porous materials have been developed to design perfect and broadband absorption[63]. Other configurations have been studied, among them the periodic partitioning of the metaporous layer[64] or a labyrinthine channel embedded in a porous host[65], based on the idea of coiling up the space of the channel as was done for non-porous metamaterial designs. Recently, a hierarchical multilayer metaporous structure was proposed using an optimization procedure, giving rise to gradient and periodic designs[66].

Some studies involving porous layers embedding periodic arrays of inclusions were done during the last 15 years[67–70]. These inclusions can have either resonances or not and they allow for an increased dissipation in the viscous regime. These structures are refered to as metaporous surfaces when used in reflection problems. The first works involving metaporous surfaces or interfaces were developed for transmission problems[71,72]. In case of more than one inclusion per spatial period, several absorption peaks can be obtained for a particular arrangement along the porous thickness[68]. Nevertheless, this rapidly leads to a thick structure. Near the inclusion resonance frequency, chosen to be in the viscous regime, an absorption peak occurs[73]. The resonance frequency of the inclusions depends on the orientation of the resonators relative to the rigid boundary[69,73]. Recently, 3D printed porous materials with embedded split-ring resonators have been modeled[61], designed and experimentally characterized showing perfect absorption peaks. More complex spiral-shaped slits have been embedded in a porous layer and have shown to be able to maintain a high absorption for large incidence angles[74]. Xiong *et al.* have studied a metaporous liner with embedded resonator and shown the existence of an exceptional point when the geometry is tuned properly, leading to enhanced absorption[75].

Previous metaporous layers studies have considered a host medium modelled as a rigid-frame porous layer; extensions of these structures to consider poroelastic materials have been done[76]. Interesting properties were exhibited for thin elastic shell inclusions, and particularly in the case of a coupling between the trapped and the first volume mode leading to a high wide band absorption. While this previous study implements a semi-analytical model that makes use of Multiple Scattering Theory (MST), it is worth noting here that different modelings exists as well, based on a Finite Elements Method (FEM) and Multiple Scattering Theory (MST) have been validated for the case of an air-saturated metaporoelastic configuration with a solid inclusion and rubber coatings[77]. This thesis is especially focused on the computation of the dispersion relation of such configurations. Therefore, a dedicated literature review dealing with the tools available to achieve such computation will be given in Chapter 1.

Fig. I.3: Figure de biblio métamatériaux, montrer les résultats utiles. qq configs genre weisser, cavalieri, li et ?

Motivations & Work Objectives

Context and Motivation This thesis is dedicated to the numerical modeling of wave propagation in complex metaporous and poroelastic multilayered structures. These materials combine the dissipative characteristics of porous media with geometric or structural features typical of acoustic metamaterials, offering advanced control over sound transmission, reflection, and absorption.

Understanding and predicting the acoustic behavior of such systems is essential for designing efficient sound absorbers, acoustic coatings, and noise-reducing structures. In particular, this work focuses on the role of guided and leaky wave modes in shaping scattering properties, and on the development of efficient numerical strategie

— precise description of the scattering properties and acoustic responses of these structures

— also studied the zeros of the scattering matrix as seen in the previous section.

— wavenumber behaviour remains unknown as of now. There is an interest of calculating the band structure or dispersion relation of metaporous layer

— help to highlight the modal behaviour responsible for these absorption peaks. Also highlight the full behaviour, might be able to exploit other phenomena.

— These studies have been limited to close systems and the impact of the surrounding radiating media should be significant on the dispersion relation heavily modifying the boundary conditions.

— Benchmark of existing numerical methods able to compute of the acoustic response of metaporous structures

— Démarches, etc. focus on the numerical aspects mostly.

Contents of the thesis

This introductory section has set the context as well as an overview of the literature in which this work is situated. We saw that acoustic metamaterials have emerged in the last decades and propose noise mitigation structures that are much more efficient that the traditional foams used for sound insulation. In particular, metaporous surfaces have shown to be promising options achieving one or multiple unitary absorption peaks at subwavelength frequency. Overall, a focus area has been identified towards identifying more precisely the modal behaviour of these configurations. Thus, in this thesis the main interest and final is to succesfully compute the dispersion relations of guided waves in such structures. The manuscript is split into four different chapters as follows,

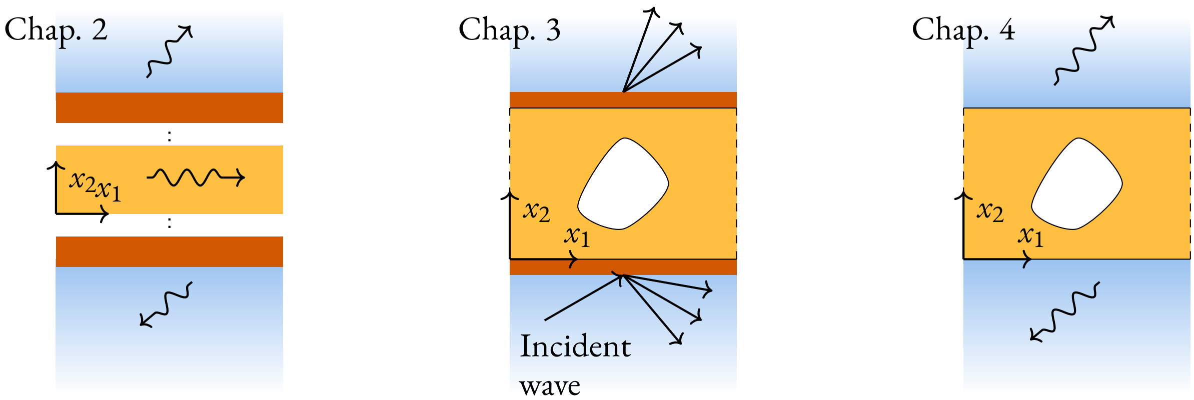

Fig. I.4: Diagram presenting the various configurations studied in this work

— Chapter 1 introduces the fundamental concepts necessary for the good understanding of the thesis. This chapter focuses on the scattering of waves on dissipative layers and the guided waves formalism that is used for the calculation of dispersion relations. Additional bibliography is proposed on these topics as well as a summary of usable numerical methods to compute such problems.

— Chapter 2 targets the study of the propagation of guided waves in multilayer arbitrary structures with dissipative layers. A simple numerical method, the Spectral Collocation Method (SCM) is implemented and extra steps allow for the local coupling of surrounding media allows for the computation of the dispersion relation for these configurations. This chapter is constituted from paper[P1].

— Chapter 3 is focused on several numerical strategies for the modelling of scattering properties of metaporous/metaporoelastic structures. A global method is coupled to a FEM. Results focused on the performance of the proposed method, compared to other similar frameworks is proposed. The convergence of this method is shown for various metaporoelastic surfaces, which geometries were studied in previous papers. This chapter is originally an article referenced as paper. cite this when finished.

— In Chapter 4, the modal behaviour of metaporous structure is studied. The fem method from the previous chapter is adapted to describe this problem. Results are presented for the case of a rigid inclusion, as well as with embedded Split-Ring Resonator (SRR). Contents for this chapter were published in paper. cite this when finished.

Finally, a summary of this work is proposed along with some perspectives on this work, including early results generalizing results from chapter four to include poroelastic hosts.

Chapter 1

Introductory concepts on guided waves, overview on numerical methods

Contents

This introductory chapter lays the foundation for the thesis, giving some notions of the theoretical aspects in order to guide the reader through the various chapters that will follow. At first, a state of the art of the literature is proposed, covering the topics of guided waves in dissipative layers, leaky modes in structures, metaporous configurations, ...

The present chapter is intended as an introduction to the fundamental concepts deemed necessary for the good understanding of the manuscript. Some theoretical aspects will be presented as well as some bibliography elements, which are useful for the context the present research takes place in.

1.1 Modelling of porous media

Porous materials are constituted of two distinct domains: an elastic matrix Ω_s with a pore network Ω_f, which is filled by some saturating fluid, they are biphasic. A diagram of their structure is shown in Fig. 1.1. The porosity is defined as the fraction of fluid over the total volume of the material, as $\phi = V(\Omega_f)/V(\Omega)$. In the limit case, $\phi = 0$ the material becomes purely a solid while at $\phi = 1$, it becomes a fluid. The geometry of the fluid phase forms a complex microstructure at a microscopic level. Fortunately, at acoustic scales, these materials can often be considered at a meso-/macroscopic scale and therefore models have been developed to describe the large-scale motion of these materials. In this context, they can be regarded as homogeneous and described by a set of effective parameters that account for their complex inner geometry. The porous structure of these materials causes the incoming acoustic energy to be dissipated through friction (viscous losses), thermal exchanges and elastic dissipation when the skeleton and the fluid are coupled. These losses are varying with the change of the porosity, pore size and geometry, and mechanical parameters among others [cite something].

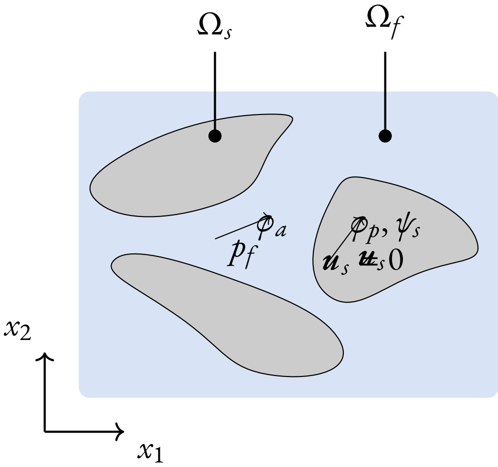

Fig. 1.1: Porous material structure

1.1.1. Rigid-frame approximation

In the so-called rigid-frame case, the skeleton of the porous material is motionless. Therefore, the pore walls are rigid walls and the material can be modeled as an *equivalent fluid* which possesses complex effective properties in order to model the losses mechanisms. Acoustic propagation is governed by the Helmholtz equation,

$$(\nabla^2 + k_{eq}^2)p(\boldsymbol{x}) = 0 \tag{1.1}$$

In the case where porous or fibrous materials have a porosity close to 1 and the frequency range of interest is limited, the Delany-Bazley model was originally developed in 1970 to propose empirical expressions for the values of the complex wavenumber and characteristic impedances[11,78] and later improved by Miki[79]. Although limited in accuracy and validity domain, these models has been widely used since it only required the flow resistivity of the porous material. Later on, a more precise semi-phenomenological model additionnaly accounting for the viscous and inertial dissipative effects was developed from 1987 to 1991. It is the Johnson-Champoux-Allard (JCA) model which models the complex effective density $\rho_{eq}(\omega)$ and bulk modulus of the material as,

$$\begin{aligned}
\rho_{\mathrm{eq}}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi}\left(1 + \frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty\omega}\sqrt{1 - \mathrm{i}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda}\right)^2}\right), \\
K_{\mathrm{eq}} &= \frac{\kappa P_0}{\phi}\left(\kappa - (\kappa-1)\middle/\left(1 + \frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty\mathrm{Pr}\omega}\sqrt{1 - \mathrm{i}\mathrm{Pr}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda'}\right)^2}\right)\right)^{-1},
\end{aligned} \tag{1.2}$$

+biot frequency.

1.1.2. Elastic motion of the skeleton: Biot theory with various formulation

While the equivalent fluid formalism is able to accurately model porous material, the rigid-wall assumption remains limited in cases where of a heavy saturating fluid, e.g. coupling between an exterior fluid and the pore fluid or when the acoustic radiation has a wavelength comparable to the pore scale. In this case, the full elastic motion should be modeled, so that to account for the coupling between the solid matrix and its fluid filling. The Biot theory[80–83]is a formalism that encompasses the theory of the deformation of porous material saturated by an incompressible fluid, used both in geophysics and acoustics. For sake of conciseness, this model won't be derived in this manuscript since it is out of scope and has been done in a few textbooks[11].

- elastic motion accounted using Biot theory + passage rapide d'une à l'autre formulation
- us, uf -> us, p
- us, uf -> us, ut
- dissipation mechanisms in porous layers
- focus on the attenuation of the waves

1.2 Wave propagation elements

1.2.1. Scattering from a porous layer

- Reflection and transmission coefficient of a porous layer ? should involve some numerics

1.2.2. Wave propagation in homogeneous layers

- guided waves formalism
- Lamb modes
- leaky modes, energy radiation: relevance of the imposition of a local coupling contrary to a full modelling of the surrounding media with some termination condition satisfying the Sommerfeld condition, PML or analogous.

1.3 Wave propagation in periodic structures

- periodic boundary conditions and periodic fields
- brillouin zone
- bragg bandgaps

1.4 Numerical tools for the modeling of the various problem types

1.4.1. Plane wave modelling

- ok for homogeneous layers
- complicated to solve dispersion relations
- requires MST for complex geometries

1.4.2. TMM and variations

- easy to implement
- but unstable :(

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1.4.3. Numerics for homogeneous layers

- 1D discretization: SCM, SAFE, etc.

1.4.4. FEM for more complex geometries

- FEM, discretization of surrounding media.

Brouillon à partir de là

thèses à mentionner: natacha aberkane-gauthier, théo cavalieri, magliacano, mathieu gaborit,

CHAPTER 2

Template d'article en tant que chapitre

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THIS CHAPTER IS BASED ON THE MATERIAL PUBLISHED IN REF. MINOR MODIFICATIONS TO THE TEXT AND FIGURES HAVE BEEN MADE TO FIT THE FORMAT AND NOTATION OF THIS THESIS.

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CHAPTER 3

A general spectral collocation method for computing the dispersion relations of guided acoustic waves in multilayer dissipative structures

Contents

Authored by: *Mathieu Maréchal, Alan Geslain, Jean-Philippe Groby, Vicent Romero-García and Olivier Dazel*.

THIS CHAPTER IS BASED ON THE MATERIAL PUBLISHED IN[P1]. MINOR MODIFICATIONS TO THE TEXT AND FIGURES HAVE BEEN MADE TO FIT THE FORMAT AND NOTATION OF THIS THESIS.

3.1 Abstract

A spectral collocation method is proposed to compute the complex wavenumber-real frequency dispersion relations of guided acoustic waves in multilayer structures involving dissipative materials. The nature of these dissipative materials is initially considered to be arbitrary, i.e., poroelastic, viscoelastic, or viscoacoustic. For a given frequency, the complex wavenumbers as well as the physical fields, which are further used to evaluate the Poynting vectors and analyze the energy flux, are

obtained by solving a generalized eigenvalue problem. The latter arises from a set of discretized equations of motion and appropriate boundary (coupling) conditions. These equations of motion and boundary (coupling) conditions are imposed by the nature of the material composing each layer of the structure. A focus is made on poroelastic layers. The dispersion relation of a two-layer elastic-poroelastic structure is analyzed, as well as the energy flows in the structure. The results as calculated with the present spectral collocation method are validated against those obtained with a classical complex root-finding (Müller) method and experiments.

3.2 Introduction

Dealing with guided acoustic waves in multilayer stuctures of dissipative (viscoelastic, viscothermal, or poroelastic) material requires the evaluation of complex wavenumber-real frequency dispersion relations. Solving for these dispersion relations consists in finding complex roots of complex characteristic equations, which often arise from complex matrix determinants and can rarely be solved analytically. Various methods[85,86], among which the Müller algorithm[87], are commonly used to achieve this challenging task. These methods usually rely on preliminary semi-analytical derivations, require multiple initial guesses, and are iterative, so they can get stuck in local minima. Computing the complex wavenumber-real frequency dispersion relation in a reliable and robust way is thus of primary importance. Commercial software such as Disperse[88] or open-source software such as Dispersion Calculator[89,90] are available for calculating dispersion curves. They use mode tracing or root-finding methods and focus mainly on multilayer elastic structures. Alternatively, several numerical methods can be used, such as the Finite Element Method[91], the Semi-Analytical Finite Element method (SAFE)[92–94], or spectral methods[95]. Spectral methods are especially relevant in this framework. They are numerical approximation techniques that seek the solution of a differential equation using a finite series of infinitely differentiable basis functions with *spectral accuracy*[96].

The propagation and dispersion of waves in multilayer systems have long been investigated[97,98]. Spectral methods have been developed to cope with wave radiation problems, but have also emerged in the last decades to solve dispersion relations[99]. Effectively, the numerical scheme boils down to discretizing the geometry along the layer thickness and solving an eigenvalue problem. Here, a collocation method is used to discretize the geometry. A finite-dimension polynomial basis is chosen and the collocation points are distributed according to the roots of this polynomial basis. Leaky modes appear in practice when multilayer structures are surrounded by one or two half-spaces. Accounting for this phenomenon brings us closer to practical considerations regarding these systems, as they would radiate part of the acoustic energy back to the surrounding medium, although it adds hardships to the modeling of the system; the waves modeled in these half-spaces should satisfy some radiation condition. These can be imposed by implementing perfectly matched layers[100], by using a bounded mapping, and applying a Spectral Collocation Method (SCM) to all layers of the extended system[101], or by using a suitable change of variable and solving a non-linear eigenvalue problem[102]. This last method inspires the current work. We propose a SCM suitable for solving complex wavenumber-real frequency dispersion relations in multilayer systems involving dissipative media. As a way of example, we will focus on multilayer poroelastic systems. Mechanical wave propagation in bulk poroelastic materials are well understood and modeled thanks to the seminal contributions of M.A. Biot[82,83,103,104] and subsequent contributions, notably by D.L. Johnson *et al.*[105] and J.-F. Allard *et al.*[19,106] concerning the modeling of viscothermal losses. Although poroelastic materials are bounded in practice and are usually encapsulated in more complex multilayer structures combining layers of materials of different nature in geophysics and acoustics, there are only a few studies on the dispersion relation of guided acoustic waves in poroelastic[76,107,108], porous[109,110] or anisotropic poroelastic layers[111]. The SCM has already been used to solve the dispersion relation of guided acoustic waves in elastic layered rings[112,113] or anisotropic elastic multilayer systems[114], but has

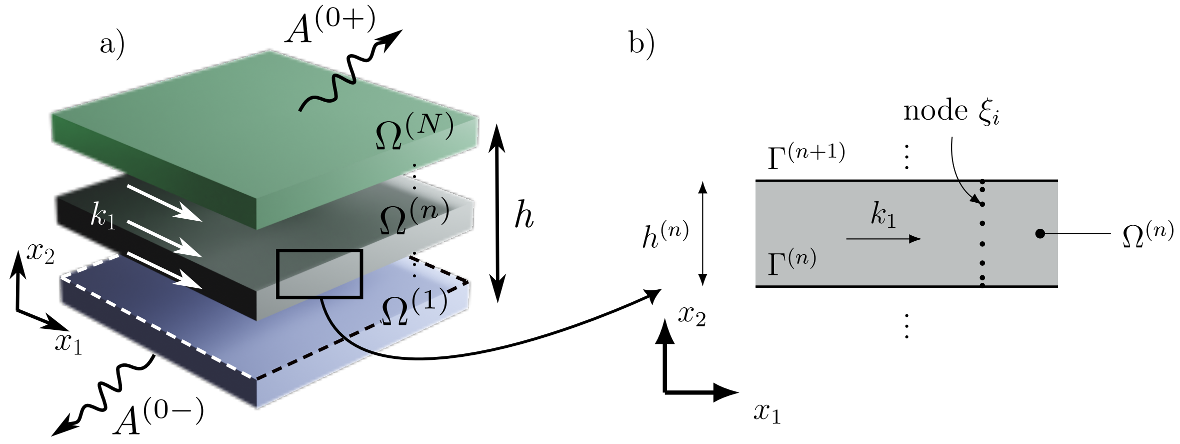

Fig. 3.1: a) Sketch of the geometry of the multilayer system and b) zoom on the n-th layer.

never been used to cope with multilayer systems involving dissipative materials of arbitrary nature.

This work aims at filling this gap and is divided into two sections. First, the SCM is derived for generic multilayer configurations possibly coupled to two surrounding identical fluid half-spaces. A two-layer elastic-poroelastic structure is then considered by way of example. The dispersion relation computed with the SCM method is compared to that obtained with a classical root-finding (Müller) method and that measured. The energy flow is further analyzed for this two-layer configuration, with very little additional computational cost. In the last section, we experimentally analyze the two-layer structure by retrieving its complex dispersion relation from displacement measurements using the SLaTCoW (Spatial LAplace Transform for COmplex Wavenumber recovery). The previously obtained numerical results can thus be correlated with the data of the dispersion relation.

3.3 Solving the dispersion relations for multilayer structures

3.3.1. Spectral Collocation Method

Figure 3.1a depicts the general geometry of the multilayer structure considered in this article. It is composed of N homogeneous layers of arbitrary nature. The n-th layer has a thickness of $h^{(n)}$ and occupies the domain $\Omega^{(n)}$. The lower and upper interfaces are denoted $\Gamma^{(n-1)}$ and $\Gamma^{(n)}$. The multilayer structure is assumed to be two-dimensional, of infinite extent along x_1, and bounded along the x_2-axis between 0 and the total structure thickness $h = \sum_{n=1}^{N} h^{(n)}$. Outgoing waves radiate above and below the structure in two fluid half-spaces $\Omega^{(0^\pm)}$, occupied by the same fluid.

We assume an implicit time dependence $e^{-\mathrm{i}\omega t}$. The physical fields associated to the guided waves in the n-th layer propagating along the positive x_1-axis take the form

$$\boldsymbol{\Theta}^{(n)}(\boldsymbol{x}) = \boldsymbol{s}^{(n)}(k_1, x_2)e^{\mathrm{i}k_1x_1}, \tag{3.1}$$

where $\boldsymbol{s}^{(n)}(k_1, x_2)$ is the spatial Laplace transform of $\boldsymbol{\Theta}^{(n)}(\boldsymbol{x})$ and k_1 is the complex valued x_1-component of the wavenumber. Note that the spatial Laplace transform is used instead of the spatial Fourier transform because k_1 is complex-valued.because k_1, the x_1 component of the wavenumber is complex-valued. The physical fields that compose $\boldsymbol{\Theta}^{(n)}(\boldsymbol{x})$ depend on the nature of the material occupying the n-th layer. Introducing $\boldsymbol{\Theta}^{(n)}(\boldsymbol{x})$ in the corresponding wave equations (of second order in space in this work) reduces to a second-order differential operator $\mathcal{L}^{(n)}$ in x_2, that depends on the material properties and can further be developed as a second-order polynomial in k_1

$$\mathcal{L}^{(n)}(x_2)\boldsymbol{s}^{(n)}(k_1,x_2) = \left(k_1^2\mathcal{L}_2^{(n)}(x_2) + k_1\mathcal{L}_1^{(n)}(x_2) + \mathcal{L}_0^{(n)}(x_2)\right)\boldsymbol{s}^{(n)}(k_1,x_2) = \boldsymbol{0}, \tag{3.2}$$

where $\mathcal{L}_j^{(n)}$ are the coefficients of the polynomial expansion $j = 0, 1, 2$. Please note that this operator is nothing but the spatial Laplace transform of the equations of motion.

The interface (boundary) conditions at $\Gamma^{(n)}$ between the $n+1$-th and the n-th layers again depend on the nature of these two layers, but can be formally written in the form

$$C^{(n)-}\ \boldsymbol{s}^{(n)}(k_1)\Big|_{\Gamma^{(n)}} - C^{(n)+}\ \boldsymbol{s}^{(n+1)}(k_1)\Big|_{\Gamma^{(n)}} = \boldsymbol{0}, \tag{3.3}$$

where $C^{(n)\pm}$ are interfaces operators, which involve differential operators of maximum second-order in x_2. Following this procedure and introducing $\boldsymbol{S} = \left(\boldsymbol{s}^{(0^-)}\dots\boldsymbol{s}^{(n)}\dots\boldsymbol{s}^{(0^+)}\right)^{\mathsf{T}}$, with $^{\mathsf{T}}$ denoting transposition, the problem can be cast in matrix form, whose coefficients are differential operators in x_2,

$$\boldsymbol{K}\boldsymbol{S}(k_1,x_2) = \begin{pmatrix} C^{(0^-)} & -C^{(0^+)} & 0 & & & & \\ 0 & \mathcal{L}^{(1)}(x_2) & 0 & & & & \\ 0 & C^{(1^-)} & -C^{(1^+)} & 0 & & & \\ & 0 & \mathcal{L}^{(2)}(x_2) & 0 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 0 & \mathcal{L}^{(N)}(x_2) & 0 \\ & & & 0 & C^{(N^-)} & -C^{(N^+)} \end{pmatrix}\boldsymbol{S}(k_1,x_2) = \boldsymbol{0}. \tag{3.4}$$

Note that the equations of motion in both upper and lower half-spaces are not required in Eq.(3.4). This system only depends on the values of the two spatial Laplace transforms $s^{(0^\pm)}$ at the upper and lower interfaces. Instead, we specify the nature of the two half-spaces, i.e. a fluid medium, and make use of the Sommerfeld condition, to explicitly give the form of the spatial Laplace transforms $p^{(0^\pm)}(k_1,x_2)$ of the pressure fields $p^{(0^\pm)}(\boldsymbol{x})$

$$\begin{aligned} p^{(0^-)}(k_1,x_2) &= A^{(0-)}e^{-\mathrm{i}k_2^{(0)}x_2}, \in \Omega^{(0^-)}, \\ p^{(0^+)}(k_1,x_2) &= A^{(0+)}e^{\mathrm{i}k_2^{(0)}(x_2-h)}, \in \Omega^{(0^+)}, \end{aligned} \tag{3.5}$$

with $k_2^{(0)} = \sqrt{\left(k^{(0)}\right)^2 - k_1^2}$, such that $\mathrm{Re}\left(k_2^{(0)}\right) \geq 0$, where $k^{(0)} = \omega/c_f$ is the wavenumber in the fluid and c_f the wave velocity. Thus, $\boldsymbol{s}^{(0^\pm)}$ appearing in Eq.(3.4) reduces to $s^{(0^\pm)} = A^{(0\pm)}$. $C^{(0^-)}$ and $C^{(N^+)}$ are therefore no longer operators and reduce to $C^{(0^-)} = \boldsymbol{C}^{(0^-)}$ and $C^{(N^+)} = \boldsymbol{C}^{(N^+)}$. In our fluid case, the interface conditions at $\Gamma^{(0)}$ and $\Gamma^{(N)}$ involve the spatial Laplace transform of the pressure and its first spatial derivative with respect to x_2 whatever the nature of the 1-st and N-th layers. Both matrices $\boldsymbol{C}^{(j)}$, $j = (0^-), (N^+)$, can thus be formally written as

$$\boldsymbol{C}^{(j)} = k_1\boldsymbol{C}_1^{(j)} + \boldsymbol{C}_0^{(j)} + k_2^{(0)}\boldsymbol{C}_{1'}^{(j)}, \tag{3.6}$$

with the subscript denoting the submatrices rearranged according to the power order $k1$. Index $1'$ corresponds to the term in $k_2^{(0)}$. Introducing Eqs.(3.2) and (3.6) in Eq. (3.4) leads to

$$\left(k_1^2\boldsymbol{K}_2 + k_1\boldsymbol{K}_1 + \boldsymbol{K}_0 + \mathrm{i}k_2^{(0)}\boldsymbol{K}_{1'}\right)\boldsymbol{S} = \boldsymbol{0}. \tag{3.7}$$

This eigenvalue problem is non-linear because of the presence of the term $k_2^{(0)}$. Making use of the following changes of variable $k_1 = k^{(0)}\left(\gamma + \gamma^{-1}\right)/2$ and $k_2^{(0)} = k^{(0)}\left(\gamma - \gamma^{-1}\right)/2i$[115,116] and companion linearization[117], a generalized eigenvalue problem is formed and solved for γ, from which the dispersion relation can be calculated. More details on this procedure can be found in[102]. Note that the first or the last column of Eq.(3.4) vanishes when radiation is absent on either side of the multilayer system. The situation is different when both fluid half-spaces disappear. The eigenvalue problem Eq.(3.7) is no longer non-linear and the problem can directly be solved after companion linearization. Note also that fluid half-spaces are only considered for the sake of simplicity and that this procedure can be adapted to any kind of material occupying the half-spaces, as described recently[118]. Note finally that this procedure avoids discretization of these two half-spaces.

To numerically solve the problem, the spatial Laplace transform of the physical fields in each n-th layer are expanded as

$$\boldsymbol{s}^{(n)}(x_2) \approx \underline{s}^{(n)}(x_2) = \sum_{m=0}^{M} \alpha_m^{(n)} \underline{\psi}_m(\xi^{(n)}), \tag{3.8}$$

where α_m are coefficients of the polynomial expansion, $\underline{\psi}_m$ are the Chebyshev polynomial of order m, and $\xi^{(n)} = 2(x_2 - \sum_{j=1}^{n-1} h^{(j)})/h^{(n)} - 1, \ \in [-1,1]$. Underlined variables represent discrete vectors and double-underlined variables discrete matrices. The goal of this approximation is to discretize the differential operators in x_2 appearing in the problem. To do so, the SCM is employed. $\xi^{(n)}$ is discretized on the roots of the Chebyshev polynomials, $\underline{\xi}_j^{(n)} = \cos\left(\frac{j\pi}{M^{(n)}}\right)$ with $j = 0 \ldots M^{(n)}$. Please note that $\underline{\xi}_j^{(n)}$ discretely runs from 1 to -1 with increasing j, while $\xi^{(n)}$ continuously runs over the same interval but in the opposite direction, i.e. from -1 to 1. These collocation points/nodes form a non-uniform grid along the thickness of each layer. This discretization is usually preferred because the collocation points are clustered at the interfaces[96]. The locations of these nodes are presented in Fig. 3.1b for the n-th layer with an arbitrary value $M^{(n)} = 8$. The differential operators in the n-th layer are thus represented by differentiation matrices (DMs), such that

$$\begin{gathered} \boldsymbol{s}^{(n)}(x_2) \rightarrow \underline{\underline{I}}{:}\underline{s}^{(n)}(\underline{\xi}_j), \qquad \partial_2 \boldsymbol{s}^{(n)}(x_2) \rightarrow \left(\frac{2}{h^{(n)}}\right) \underline{\underline{D_2}}{:}\underline{s}^{(n)}(\underline{\xi}_j) = \underline{\underline{D_2}}^{(n)}{:}\underline{s}^{(n)}(\underline{\xi}_j), \\ \partial_{22} \boldsymbol{s}^{(n)}(x_2) \rightarrow \left(\frac{2}{h^{(n)}}\right)^2 \underline{\underline{D}}_{22}{:}\underline{s}^{(n)}(\underline{\xi}_j) = \underline{\underline{D}}_{22}^{(n)}{:}\underline{s}^{(n)}(\underline{\xi}_j). \end{gathered} \tag{3.9}$$

where $\underline{\underline{I}}$ is the identity matrix and $\underline{\underline{D}}_2$ and $\underline{\underline{D}}_{22}$ are respectively the first-order and second-order normalized DMs. Contrary to an interpolation with Chebyshev polynomials, DMs interpolate the functions and their spatial derivatives directly and thus do not involve the polynomial expansion coefficients per se.[119]. Note that the normalized DMs are denoted as $\underline{\underline{D}}_2^{(n)}$ and $\underline{\underline{D}}_{22}^{(n)}$ in the following . These matrices are square and have a size of $(M \times M)$. This size also directly depends on the order of the considered Chebyshev polynomials. The formulation prevents the occurrence of roundoff errors, which is of particular importance when SCM is applied to solve eigenvalue problems[120].

The linear operator $\mathcal{L}^{(n)}$ is discretized as a matrix $\underline{\underline{L}}^{(n)}$ with size $(P^{(n)}(M^{(n)}-1), P^{(n)}(M^{(n)}+1))$, where $P^{(n)}$ is the number of physical fields required to describe the wave propagation in the layer and again depends on the nature of the material this layer is composed of. The first and the last rows of the DMs are removed because they are used to apply the interface conditions at $\Gamma^{(n+1)}$ and $\Gamma^{(n)}$ respectively. The interface operators $C^{(n)\pm}$ are effectively expressed in terms of this first row of the DMs of the n-th layer and this last row of the DMs of the $(n+1)$-th layer. These rows respectively correspond to the first, i.e. $\underline{\xi}_{M^{(n)}} = -1$ and $\xi^{(n)} = 1$, and last collocation nodes of the n-th and $(n+1)$-th layers, i.e. $\underline{\xi}_0 = 1$ and $\xi^{(n+1)} = -1$. These vectors are subsequently denoted $\underline{I}^{(n+1)-}$, $\underline{I}^{(n)+}$, $\underline{D}_2^{(n+1)-}$, $\underline{D}_2^{(n)+}$, $\underline{D}_{22}^{(n+1)-}$, and $\underline{D}_{22}^{(n)+}$. The interface matrix $\underline{\underline{C}}^{(n)^-}$ is of size $(P^{(n)} + P^{(n+1)}, P^{(n)}(M^{(n)} + 1))$ and the interface matrix $\underline{\underline{C}}^{(n)^+}$ is of size $(P^{(n)} + P^{(n+1)}, P^{(n+1)}(M^{(n+1)} + 1))$.

Finally, the full $(\sum_{n=0}^{N} M^{(n)}P^{(n)}+2, \sum_{n=0}^{N} M^{(n)}P^{(n)}+2)$-matrix system for the general multilayer system is

$$\underline{\underline{K}}:\underline{S} = \begin{pmatrix} \underline{\underline{C}}^{(0^-)} & -\underline{\underline{C}}^{(0^+)} & 0 & & & & \\ 0 & \underline{\underline{L}}^{(1)} & 0 & & & & \\ 0 & \underline{\underline{C}}^{(1^-)} & -\underline{\underline{C}}^{(1^+)} & 0 & & & \\ & 0 & \underline{\underline{L}}^{(2)} & 0 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 0 & \underline{\underline{L}}^{(N)} & 0 \\ & & & 0 & \underline{\underline{C}}^{(N^-)} & -\underline{\underline{C}}^{(N^+)} \end{pmatrix} \begin{pmatrix} A^{(0-)} \\ \underline{s}^{(1)} \\ \underline{s}^{(2)} \\ \vdots \\ \underline{s}^{(N)} \\ A^{(0+)} \end{pmatrix} = \underline{0}. \tag{3.10}$$

This system is cast in the form of a generalized eigenvalue problem that is solved by traditional eigenvalue solvers at each frequency. The dispersion relation for the acoustic guided wave in the dissipative multilayer system is calculated, repeating the procedure for each frequency ω.

3.3.2. Solving the dispersion relation with Müller algorithm

The dispersion relation of multi-layer dissipative structure are more commonly calculated with secant-based algorithm as the Müller algorithm. The latter relies on a semi-analytic description of the fields. These fields are best written in each layer, independently of the nature of the material it is composed of, using the potentials, i.e. decomposing the field $\theta = \nabla\varphi + \nabla\times\psi$ into an irrotational component φ and a transverse component $\psi = \psi e_3$. The spatial Laplace transform of the potentials in the n-th layer are entirely described by

$$\tilde{\theta}_n(k_1, x_2) = A_X^{(n)+} e^{\mathrm{i}k_{2,X}^{(n)}(x_2-h^{(n)})} + A_X^{(n)-} e^{-\mathrm{i}k_{2,X}^{(n)}(x_2-h^{(n)})}, \tag{3.11}$$

with $A_X^{(n)\pm}$ the up and down-going wave amplitudes, $k_{2,X}^{(n)} = \sqrt{\left(k_X^{(n)}\right)^2 - k_1^2}$, such that $\mathrm{Re}\left(k_{2,X}^{(n)}\right) \geq 0$, and X refers to the type of wave, i.e. shear, compressional or acoustic. If necessary, these Laplace transform representations are complemented by those of the pressure field in the fluid half-spaces provided in Eq.(3.5).

The boundary conditions are applied at each interface and the set of equations is cast into the following matrix form $\boldsymbol{M}(k_1)\boldsymbol{q} = 0$, with $\boldsymbol{q}$ a vector containing the amplitudes of each field/potential. The modes of the system correspond to the wavenumbers k_1 for which the matrix $\boldsymbol{M}(k_1)$ is singular, i.e.

$$\det(\boldsymbol{M}(k_1)) = 0. \tag{3.12}$$

These wavenumbers are the complex roots of an implicit complex-valued equation. They are evaluated by running our own implementation of the Müller algorithm[87] from 3 initial guesses. This equation is solved iteratively for each frequency ω, giving the dispersion relation.

In addition, $\boldsymbol{q}$ can be obtained by computing the nullspace of $\boldsymbol{M}$[121] (This null space is computed using singular value decomposition as embedded in the `linalg.null_space` function from the Python package SciPy). The amplitudes of the potentials in the layers are thus evaluated, enabling all the physical fields that depend on these potentials to be calculated. However, the $\boldsymbol{M}$ matrix becomes increasingly tedious to write as the number of layers increases, and even more so when it comes to calculating the roots of its determinant and nullspaces. Nevertheless, this approach remains a fairly effective way of providing a reference solution for validating numerical results when studying a two-layer system.

3.4 Application to a two-layer poroelastic-elastic structure

In this section, the calculation of the dispersion relation for guided waves in a two-layer structure, i.e. $N = 2$, is used as an example of the application of the present SCM. This two-layer structure consists of a $h^{(1)}$-thick poroelastic layer, coated on one side by a $h^{(2)}$-thick aluminum plate, that is radiating in a fluid half-space, and rigidly backed on the other side as depicted in Fig. 3.2a. The SCM results are analyzed and compared to those obtained with the Müller root-finding method.

3.4.1. Numerical scheme

The pressure field $p^{(1)}$ and the two components of the frame displacement, $u_{1,s}^{(1)}$ and $u_{2,s}^{(1)}$, i.e. $P^{(1)} = 3$, as well as the two-components of the elastic displacement, $u_1^{(2)}$ and $u_2^{(2)}$, i.e. $P^{(2)} = 2$, are used to model the propagation in the poroelastic and in the elastic layers respectively. The $\{\mathbf{u}_s, p\}$ formulation is used[122] for the poroelastic medium. The more usual displacement formulations ($\{\mathbf{u}_s, \mathbf{u}_f\}$[82] or $\{\mathbf{u}_s, \mathbf{w}\}$[103]) cannot be used in the present case, because the number of physical fields they rely on, i.e. 4, is different from that of the potentials in the layer, i.e. 3.

In total, 5 physical fields inside the two-layer structure and an additional amplitude for the acoustic wave radiated towards $x_2 \to \infty$ are needed to solve for the dispersion relation. Each domain is discretized on $M^{(j)}$, $j = 1, 2$, collocation points.

In more details, the $\{\boldsymbol{u}_s, p\}$ formulation of the Biot theory[122] is employed to model the propagation in the homogeneous isotropic poroelastic layer. The coupled equations of motion read as

$$\begin{cases} \nabla \cdot \hat{\boldsymbol{\sigma}}_s^{(1)} + \hat{\rho}\omega^2 \boldsymbol{u}_s^{(1)} + \beta \boldsymbol{\nabla} p^{(1)} = 0, \\ \dfrac{\nabla^2 p^{(1)}}{\rho_{22}\omega^2} - \dfrac{\beta}{\phi^2} \nabla \cdot \boldsymbol{u}_s^{(1)} + \dfrac{1}{R} p^{(1)} = 0, \end{cases} \tag{3.13}$$

where $\hat{\boldsymbol{\sigma}}_s^{(1)} = \hat{A}\nabla \cdot \boldsymbol{u}_s^{(1)} \boldsymbol{I} + 2N_s \boldsymbol{\epsilon}_s$ is the *in vacuo* solid stress tensor, with $\boldsymbol{\epsilon}_s$, the solid-phase strain tensor. The parameters are generally expressed in terms of the elastic coefficients P, Q and R and the effective densities ρ_{11}, ρ_{12} and ρ_{22} initially introduced by Biot[11,82] as

$$\hat{A} = P - 2N_s - \frac{Q^2}{R}, \quad \hat{\rho} = \rho_{11} - \frac{\rho_{12}^2}{\rho_{22}}, \quad \beta = \phi\left(\frac{\rho_{12}}{\rho_{22}} - \frac{Q}{R}\right). \tag{3.14}$$

These parameters depend on the frame properties and on the effective density $\rho_{eq}(\omega)$ and bulk modulus $K_{eq}(\omega)$ of the fluid phase as reminded in 3.6.A. In short, a poroelastic material is entirely described by the porosity ϕ, the shear modulus of the frame N_s, the Poisson ratio ν, the density of the frame ρ_1, the viscous and thermal characteristic lengths Λ and Λ', tortuosity α_∞, and the flow resistivity R_f when saturated by a light fluid[82,105,106]. All these parameters are real valued except for the shear modulus, which incorporates a viscoelastic damping factor.

Introducing $\boldsymbol{s}^{(1)} = (\pi_1^s, \pi_2^s, \hat{p})^\top$, the spatial Laplace transform of Eq. (3.13) is expanded in the form of Eq. (3.2) with

$$\begin{aligned} \mathcal{L}_0^{(1)} &= \begin{pmatrix} N_s\partial_{22} + \hat{\rho}\omega^2 & 0 & 0 \\ 0 & \hat{P}\partial_{22} + \hat{\rho}\omega^2 & \beta\partial_2 \\ 0 & -\beta/\phi^2\partial_2 & 1/R + \partial_{22}/(\rho_{22}\omega^2) \end{pmatrix}, \\ \mathcal{L}_1^{(1)} &= \begin{pmatrix} 0 & \mathrm{i}\hat{P}\partial_2 & \mathrm{i}\beta \\ \mathrm{i}(\hat{A} + N_s)\partial_2 & 0 & 0 \\ -\mathrm{i}\beta/\phi^2 & 0 & 0 \end{pmatrix}, \text{ and } \mathcal{L}_2^{(1)} = \begin{pmatrix} \hat{P} & 0 & 0 \\ 0 & N_s & 0 \\ 0 & 0 & 1/(\rho_{22}\omega^2) \end{pmatrix}. \end{aligned} \tag{3.15}$$

Finally, the discretized equations of motion read as

$$\underline{\underline{L}}^{(1)}\underline{s}^{(1)} = \begin{pmatrix} (\omega^2\tilde{\rho} - k_1^2\hat{P})\underline{\underline{I}} + N_s\underline{\underline{D}}_{22} & \mathrm{i}k_1(\hat{P} - N_s)\underline{\underline{D}}_{22} & -\mathrm{i}k_1\beta\underline{\underline{I}} \\ \mathrm{i}k_1(\hat{P} - N_s)\underline{\underline{D}}_2 & \hat{P}\underline{\underline{D}}_{22} + (\omega^2\tilde{\rho} - k_1^2 N_s)\underline{\underline{I}} & \beta\underline{\underline{D}}_2 \\ \dfrac{-\mathrm{i}\beta\omega^2}{\phi^2}\underline{\underline{I}} & \dfrac{-\beta\omega^2}{\phi^2}\underline{\underline{D}}_2 & \dfrac{k_1^2}{\rho_{22}}\underline{\underline{I}} + \dfrac{\omega^2}{R}\underline{\underline{I}} \end{pmatrix}\underline{s}^{(1)} = 0. \tag{3.16}$$

The equation of motion in the elastic medium is derived from the fundamental elasticity equations[123] and reads as,

$$\rho\omega^2\boldsymbol{u}^{(2)} + (\lambda + \mu)\boldsymbol{\nabla}(\nabla \cdot \boldsymbol{u}^{(2)}) + \mu\nabla^2\boldsymbol{u}^{(2)} = 0, \tag{3.17}$$

where ρ is the density and λ and μ are the Lamé coefficients. The corresponding discretized equation of motion is

$$\underline{\underline{L}}^{(2)}\underline{s}^{(2)} = \left(\frac{\rho^{(2)}}{\mu}\omega^2\underline{\underline{I}} + \begin{pmatrix} -k_1^2(\lambda/\mu + 2)\underline{\underline{I}} + \underline{\underline{D}}_{22} & \mathrm{i}k_1(\lambda/\mu + 1)\underline{\underline{D}}_2 \\ \mathrm{i}k_1(\lambda/\mu + 1)\underline{\underline{D}}_2 & (\lambda/\mu + 2)\underline{\underline{D}}_{22} - k_1^2\underline{\underline{I}} \end{pmatrix}\right)\underline{s}^{(2)} = 0. \tag{3.18}$$

In addition, the same procedure can be followed for the equation of motion of a wave propagating in a layer of fluid material. We give its expression in order to provide an exhaustive list of the types of layers that can be modeled using the present method. The Helmholtz equation, $(\nabla^2 + k_0^2)p = 0$, with k_0 the bulk wavenumber and p the pressure field in the fluid material, is discretized into the form,

$$\left(\underline{\underline{D}}_{22} - k_1^2 + k_0^2\right)\underline{p} = 0. \tag{3.19}$$

The interface and boundary conditions depend on the vectors $\underline{I}^{\pm}$, $\underline{D}_2^{\pm}$ and $\underline{D}_{22}^{\pm}$ which corresponds to the last (subscript +) and first row (subscript −) of the DMs. From the bottom to the top of the two-layer system, they are

— Rigid boundary condition at $\Gamma^{(0)}$ which leads to

$$\boldsymbol{u}^{(1)}(x_2 = 0) = 0; \quad \boldsymbol{u}_f^{(1)}(x_2 = 0)\cdot\boldsymbol{n} - \boldsymbol{u}_s^{(1)}(x_2 = 0)\cdot\boldsymbol{n} = 0, \tag{3.20}$$

the discretized form of which is,

$$\underline{\underline{C}}^{(0)}\underline{s}^{(1)} = \begin{pmatrix} 0 & \underline{I}^+ & 0 \\ 0 & 0 & \underline{I}^+ \\ \dfrac{\phi}{\rho_{22}}\underline{D}_2^+ & 0 & -\left(1 + \dfrac{\rho_{12}}{\rho_{22}}\right)\omega^2\underline{I}^+ \end{pmatrix}\begin{pmatrix} \underline{p}^{(1)} \\ \underline{u}_1^{(1)} \\ \underline{u}_2^{(1)} \end{pmatrix} = \underline{0}. \tag{3.21}$$

— Continuity between the poroelastic and the elastic media[124] at $\Gamma^{(1)}$ which leads to

$$\begin{gathered}\boldsymbol{\sigma}^{t(1)}(x_2 = h^{(1)}) - \boldsymbol{\sigma}^{(2)}(x_2 = h^{(1)}) = 0; \quad \boldsymbol{u}^{(2)}(x_2 = h^{(1)}) = \boldsymbol{u}_s^{(1)}(x_2 = h^{(1)}); \\ \boldsymbol{u}_f^{(1)}(x_2 = h^{(1)})\cdot\boldsymbol{n} - \boldsymbol{u}_s^{(1)}(x_2 = h^{(1)})\cdot\boldsymbol{n} = 0,\end{gathered} \tag{3.22}$$

where $\boldsymbol{\sigma}^{t(1)} = \hat{\boldsymbol{\sigma}}_s - \phi\left(1 + Q/R\right)$ is the total stress tensor in the poroelastic layer. The corresponding discretized interface matrices, arising from

$$\underline{\underline{C}}^{(1)+}\underline{s}^{(2)} - \underline{\underline{C}}^{(1)-}\underline{s}^{(1)} = \underline{\underline{C}}^{(1)}\left(\underline{p}^{(1)} \quad \underline{u}_1^{(1)} \quad \underline{u}_2^{(1)} \quad \underline{u}_1^{(2)} \quad \underline{u}_2^{(2)}\right)^{\mathsf{T}} = 0, \tag{3.23}$$

are cast as

$$\underline{\underline{C}}^{(1)} = \left(\begin{array}{ccc|cc} -N_s\underline{D}_2^+ & -\mathrm{i}k_1 N_s \underline{I} & \underline{0} & \mu\underline{D}_2^- & \mathrm{i}k_1\mu\underline{I}^- \\ -\mathrm{i}k_1\hat{A}\underline{I}^+ & -\hat{P}\underline{D}_2^+ & \phi\left(1+\dfrac{Q}{R}\right)\underline{I}^+ & \mathrm{i}k_1\lambda\underline{I}^- & (\lambda+2\mu)\underline{D}_2^- \\ \underline{0} & -\underline{I}^+ & \underline{0} & \underline{I}^- & \underline{0} \\ \underline{0} & \underline{0} & -\underline{I}^+ & \underline{0} & \underline{I}^- \\ -\dfrac{\phi}{\rho_{22}}:\underline{D}_2^+ & \left(1+\dfrac{\rho_{12}}{\rho_{22}}\right)\omega^2\underline{I}^+ & \underline{0} & \underline{0} & \underline{0} \end{array}\right), \tag{3.24}$$

with the left-hand side corresponding to $\underline{\underline{C}}^{(1)-}$ and the right-hand side to $\underline{\underline{C}}^{(1)+}$.

— Continuity between the elastic and the fluid media at $\Gamma^{(2)}$ which leads to

$$\begin{gathered} \sigma_{12}^{(2)}(x_2 = h) = 0; \qquad \sigma_{22}^{(2)}(x_2 = h) = -p^{(0+)}; \\ u_2^{(2)}(x_2 = h) = u_2^{(0+)}(x_2 = h). \end{gathered} \tag{3.25}$$

The corresponding discretized interface matrix $\underline{\underline{C}}^{(2)}\begin{pmatrix}\underline{u}_1^{(2)} & \underline{u}_2^{(2)} & A^{(0+)}\end{pmatrix}^{\mathsf{T}} = 0$ is

$$\underline{\underline{C}}^{(2)} = \left(\begin{array}{cc|c} \underline{D}_2^+ & \mathrm{i}k_1\underline{I}^+ & 0 \\ \mathrm{i}k_1\lambda\underline{I}^+ & (\lambda+2\mu)\underline{D}_2^+ & 1 \\ \underline{0} & \omega^2\underline{I}^+ & \dfrac{\mathrm{i}k_2^{(0)}}{\rho_f} \end{array}\right). \tag{3.26}$$

Combining the discretized interface conditions, Eqs.(3.21), (3.24) and, (3.26), with the discretized equations of motion, Eqs.(3.16) and (3.18), leads to the following system

$$\begin{pmatrix} \underline{\underline{C}}^{(0)} & 0 & 0 \\ \underline{\underline{L}}^{(1)} & 0 & 0 \\ \underline{\underline{C}}^{(1)-} & -\underline{\underline{C}}^{(1)+} & 0 \\ 0 & \underline{\underline{L}}^{(2)} & 0 \\ 0 & \underline{\underline{C}}^{(2)-} & -\underline{\underline{C}}^{(2)+} \end{pmatrix} \begin{pmatrix} \underline{s}^{(1)} \\ \underline{s}^{(2)} \\ A^{(0+)} \end{pmatrix} = \underline{0}. \tag{3.27}$$

This system is further expanded in the form of Eq.(3.7) and solved for a set of eigenvalues γ and associated eigenvectors $\boldsymbol{S}$ at each frequency ω. Wavenumbers k_1 are then obtained from γ. The dispersion curve is thus calculated over a given frequency range. However, the condition $\mathrm{Re}\left(k_2^0\right) \geq 0$ should be imposed to sort out the solution that do not meet it.

Although usual and simple, this two-layer structure is challenging from a numerical point of view because it involves materials with very large impedance contrasts and layers whose thicknesses differ by several orders of magnitude. These problems are partially solved by normalizing each line and rearranging the terms of the matrix Eq. (3.7), leading to a better numerical conditioning of the system.

3.4.2. Dispersion relation

The material properties of the $h^{(1)} = 52$ mm-thick poroelastic (melanime) layer are listed in Table 3.1, while the properties of the $h^{(2)} = 1$ mm-thick aluminium plate are the density $\rho = 2700$ kg.m^{-3} and the Lamé coefficients $\lambda = 60.75$ GPa and $\mu = 26.03$ GPa. The saturating fluid and

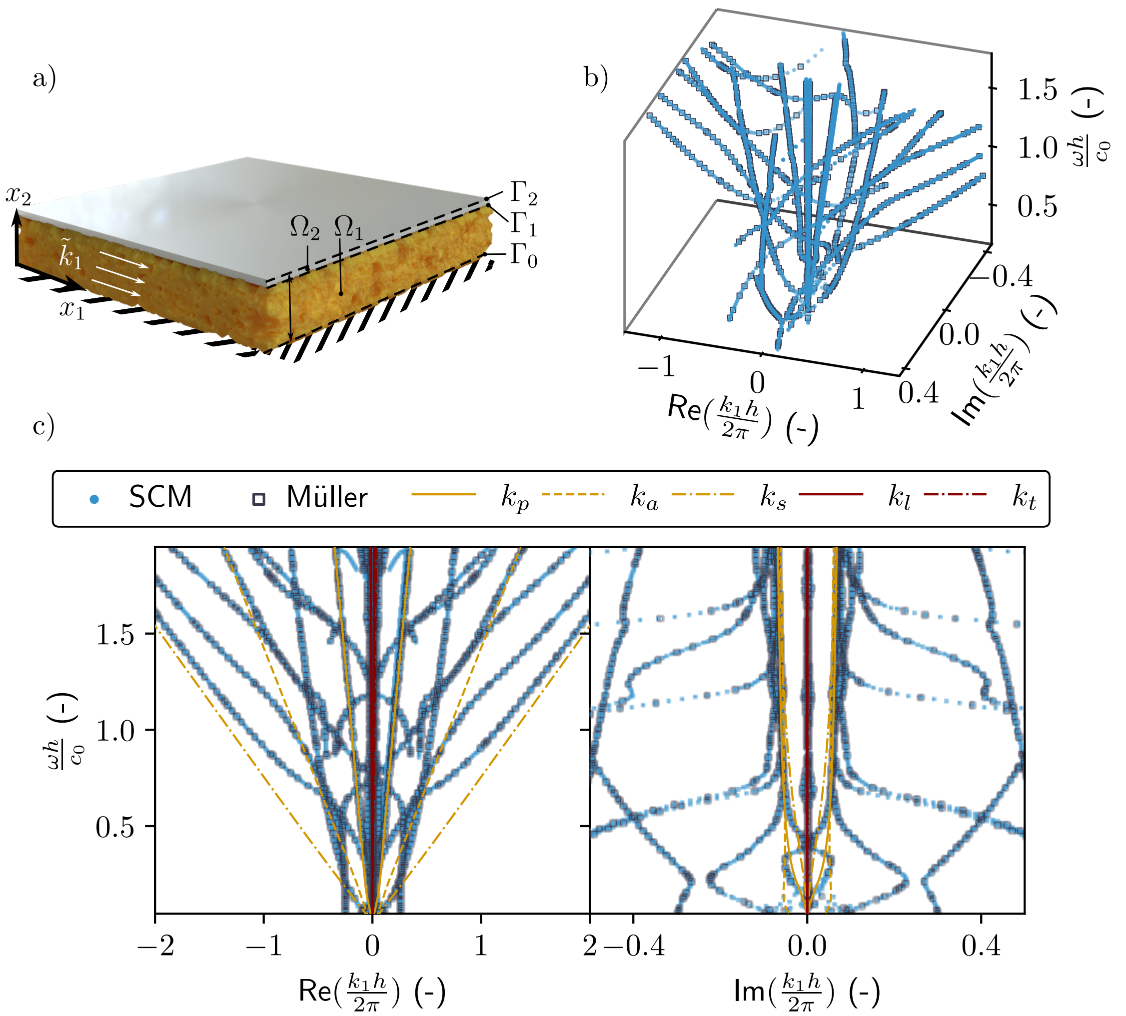

Fig. 3.2: a) Sketch of the two-layer structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. Blue dots represents the results as calculated with the SCM and black open markers represents those as calculated with the Müller method. Bulk wavenumbers in the poroelatic and elastic materials are highlighted in yellow and dark red respectively.

that occupying the half-space is air with density $\rho_f = 1.213$ kg.m^{-3}, the heat capacity ratio $\kappa = 1.4$, kinematic compressibility $\mu_f = 1.839 \times 10^{-5}$ Pa.s, the Prandtl number Pr $= 0.71$, and adiabatic bulk modulus κP_0, where the atmospheric pressure is $P_0 = 1.013 \times 10^5$ Pa. The total two-layer thickness is thus $h = h^{(1)} + h^{(2)} = 53$ mm. $M^{(1)} = 5$ and $M^{(2)} = 11$ collocation points are employed respectively in the poroelastic and elastic layers.

The 3D view of the complex dispersion diagram is depicted Fig. 3.2b. To ease readability, real and imaginary parts are depicted separately in Fig. 3.2c. Please note that this last representation can be misleading because only the modes that lie in the complex wavenumber range depicted in Fig. 3.2b are represented in Fig. 3.2c. Cut-on frequencies are thus fictitious. The results calculated using the SCM method, plotted in blue markers, are compared with those obtained using the Müller method, plotted with black open markers. Both methods provide identical results, up to around four digits, therefore validating the present approach. Note that the SCM solutions are used as initial guesses of the Müller algorithm to shorten computational time. In this way, fewer iterations are needed for the results of the root-finding method to converge. The main advantages of SCM are that no initial guesses are needed, i.e. as long as the discretization is sufficient to model the full wave behavior in the structure, the dispersion relation is guaranteed to be calculated. In addition, the matrices used for the numerical model are very small compared to those required by other methods such as FEM, and are therefore much less demanding in terms of calculation.

Table 3.1: Parameters of the poroelastic material (melamine foam)

Parameter	ϕ	ρ_1	R_f	Λ	Λ'	α_∞	ν	N_s
Unit	-	kg.m^{-3}	kPa.s.m^{-2}	μm	μm	-	-	kPa
Melamine	0.98	6.5	5.6	214	214	1	0.24	11.96 (1 + i0.07)

Wavenumbers are perfectly symmetric with both Re $(k_1 h/2\pi) = 0$, and Im $(k_1 h/2\pi) = 0$ axis, which is a feature of reciprocal systems. Wavenumbers having a very large slope and a low imaginary part correspond to modes mostly propagating in the aluminium plate. They are very close to the bulk longitudinal k_l and tranverse k_t wavenumbers in the aluminum. This is due to the very large contrast between the elastic properties of the aluminum and those of the poroelastic frame. The other branches asymptotically tend toward the bulk acoustic, k_a, compression, k_p, or shear, k_s, wavenumbers of the poroelastic medium at high frequency. At low frequencies, these modes are highly dispersive.

If we compare these results with the dispersion relation depicted in 3.6.Bc) for the guided waves in a single poroelastic plate twice as thick and in the absence of the elastic plate, we find some essential differences . The branches associated with the elastic plate and the branch clearly associated with the coupling between the elastic and poroelastic layers (referenced as (1) in Fig.3.6.Bc)) do not exist. The other branches are similar to those of this single poroelastic layer, although slightly modified by the presence of the elastic plate. This proves a weak coupling between the elastic and poroelastic layers, particularly at high frequencies. At lower frequencies, the coupling between the two layers is stronger. Effectively, the wavelength is much larger than the thickness of at least one of the two layers in these frequency ranges, which favors coupling. The A_0 branch observed in the case of a single poroelastic layer disappear in the two-layer system rigidly backed, because this rigid-backing does not allow the existence of such a mode.

3.4.3. Mode shapes and energy flow

Beyond the SCM numerical efficiency, a significant advantage of the method is that the eigenvectors associated to each eigenvalue directly provides the associated mode shape. Since these mode shapes, i.e. eigenvectors, are calculated from an eigenvalue problem, their amplitude are defined up to a constant. They are normalized to the value of the pressure field at the bottom of the poroelastic layer, i.e. at the location of the rigid backing, because this condition imposes $\boldsymbol{u}^{(1)}(0) = 0$ (and in particular $u_2^{(1)}(0) = 0$), which is equivalent to a maximum pressure field. From these normalized eigenvectors, any physical fields can be calculated as detailed in 3.6.C.

Let us consider two solutions $k_1^{\pm} = \pm|\mathrm{Re}\,(k_1)\,|$ symmetric in the dispersion diagram with respect to the axis Re $(k_1) = 0$ as shown in Figs. 3.3a-b. The displacement fields and total energy fluxes in each direction are depicted in Fig. 3.3c-f. The results calculated using the SCM method, plotted in continuous lines, match those obtained using the Müller method, plotted with marker lines. These two modes are identical but propagate in opposite directions, i.e., the mode k_1^+ propagates towards positive x_1 while the mode k_1^- propagates towards negative x_1. Displacement fields and energy fluxes are identical along the x_2 axis but are of equal modulus but opposite sign along the x_1 axis (see Fig. 3.3c-d and e-f). Both displacement field and energy flux amplitudes are two to three orders of magnitude smaller in the elastic plate (for $h_1 < x_2 < h$) than in the poroelastic plate. The aluminum coating is almost acting as a rigid boundary condition, as also testified by the fact that the x_1 components of the displacement field and energy fluxes are almost symmetric with respect to the axis $x_2 = h_1/2$, while their x_2 components are almost antisymmetric along the x_2 axis. Note that the fluid phase Poynting vectors depicted in Fig. 3.3f are of equal modulus but opposite sign, symmetric with respect to the axis $x_2 = h_1/2$, and more than a hundred times smaller than total energy flux in amplitude. The energy of these modes are thus mainly localized in the solid phase of the poroelastic plate.

To go a step further, we evaluated the energy transport velocity $\overline{V_e} = \overline{P}_t/\overline{U}$ as the ratio between the average energy flux in each layer $\overline{P}_t$ over the mean total energy $\overline{U}$. The energy transport velocity of the mode highlighted in orange in Fig. 3.3a-b is compared to the group velocity $v_g = \partial\omega/\partial\mathrm{Re}\,(k_1)$[125] in Fig. 3.3g. It is numerically computed by doing finite differences on a branch of the dispersion relation. Both velocities match for frequency ranges where the imaginary part of k_1 is low, i.e., for weak attenuation modes[126–129], but are different for strong attenuation. In particular, the group velocity diverges around the cut-on frequency. This is the reason why the energy transport velocity is usually preferred to study absorbing structures. The energy transport velocity $\overline{V_e}$ of the mode highlighted in green Fig. 3.3a-b is depicted in Fig. 3.3h. This energy transport velocity is always positive, while k_1 crosses the $\mathrm{Re}\,(k_1) = 0$ axis. When $\mathrm{Re}\,(k_1) < 0$ the phase velocity is negative, while the $\overline{V_e} > 0$. This branch (and the symmetric one) has also a negative group velocity as also testified by the fact $\mathrm{Re}\,(k_1) < 0$ while $\mathrm{Im}\,(k_1) > 0$ in the shaded area in Fig. 3.3h. This branch has also an exponentially growing amplitude.

3.4.4. Experimental validation

The experimental set-up is depicted in Fig. 3.4a. The sample is the two-layer structure considered in the previous subsection. The aluminum plate is glued on top of the poroelastic layer. The latter layer is glued on a 3 cm-thick aluminium plate, which mimic the rigid backing. The three elements are 45 cm wide and 85 cm long. The surrounding medium and the saturating fluid is air, at regular pressure and temperature conditions. The sample is positioned perpendicularly to the ground on its longest side as depicted in Fig. 3.4a to ease the mode excitation in the poroelastic layer. The excitation is provided by a shaker (Brüel and Kjaer type 4810), which is rigidly attached to the sample with a threaded steel rod fixed to the shaker on one side and glued on a 1mm thick aluminum plate of width 10 cm and height 1.5 cm. This plate is cut at the edge opposite to the threaded steel rod and glued to the porous sample, creating a line source at the edge of the sample. The resonance of this part was measured to be 4500 Hz, which is the upper limit of the measured frequency range. The excitation signal is a swept-sine function with 400 points ranging from 1 kHz to 3 kHz. The general layout of the experiments is similar to that used originally in[130]. The key difference in our case is that the excitation is operated directly to the poroelastic layer and not to the top of the aluminum plate, because the modes that corresponds to propagating solutions in the poroelastic layer are encapsulated in between two almost rigid plates (see Section 3.4.3). The normal displacement is measured with a laser Doppler vibrometer (Polytec VibroFlex Neo) on a line of length $L = 40$ cm, along the side of the thickness of the porous layer. These measurements are averaged 100 times. The excitation signal and the measured field are interfaced through a computer via a Zürich Instruments acquisition card.

The SLaTCoW method is used to analyze the space-frequency displacement measurements and recover the complex wavenumber-real frequency dispersion relation. This method relies on the Laplace spatial transform of these measurements, which gives access to the real and imaginary components of the complex wavenumber. The difference between this Laplace transform and a correctly chosen ansatz function is finally minimized for each frequency, to recover the dispersion relation. More details on this method are given in Appendix 3.6.E as well as in the original article[130]. The complex wavenumber-real frequency dispersion relation in Fig. 3.4b-c depicts the wavenumbers recovered experimentally k_s and simulated k_n. Four modes were sought during the minimization with the SLaTCow method: three of them correspond to the three bulk waves due to excitation and the fourth is depicted. The experimentally recovered mode jumps from one branch to the other because of the large modal density. The cut-on frequencies of the modes are clearly visible, especially on the imaginary part of the wavenumber. The lower real wavenumber parts of the

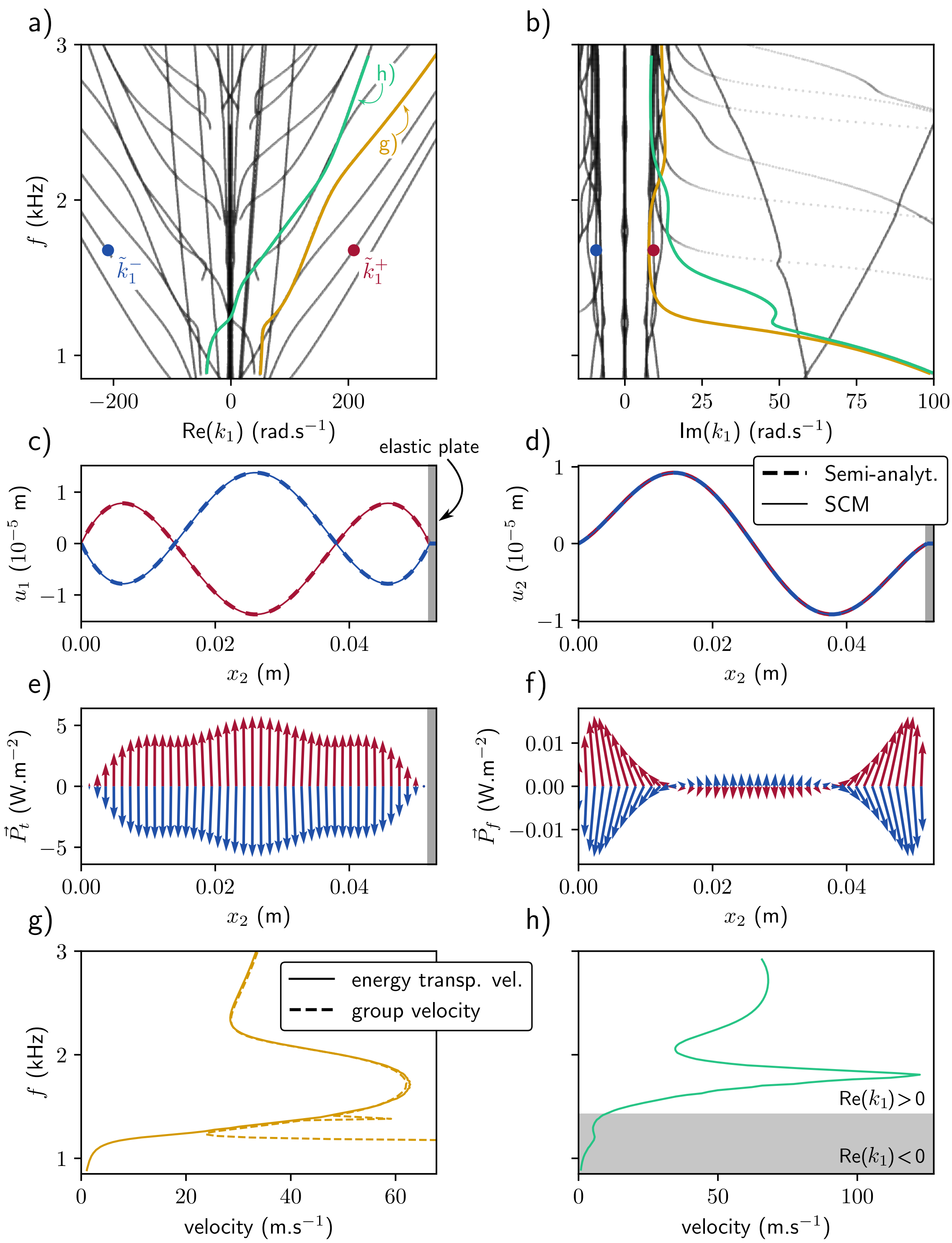

Fig. 3.3: a) and b) represent the real and imaginary parts of the dispersion relation of the two-layer structure. Two branches are highlighted in yellow and green, as well as two specific solutions, k_1^- (blue) and k_1^+ (red), symmetric with respect to the Re $(k_1) = 0$ axis. Displacement fields in the x_1 c) and x_2 direction d) associated with the k_1^+ and k_1^- solutions displayed. Total Poynting e) vector and fluid Poynting vector f) with the length of the arrow indicating the amplitude and angle indicating the direction of the vector field. g) Group and energy transport velocities for the yellow branch. h) Energy transport velocity for the green branch. The gray area represents the frequency range over which this velocity is negative.

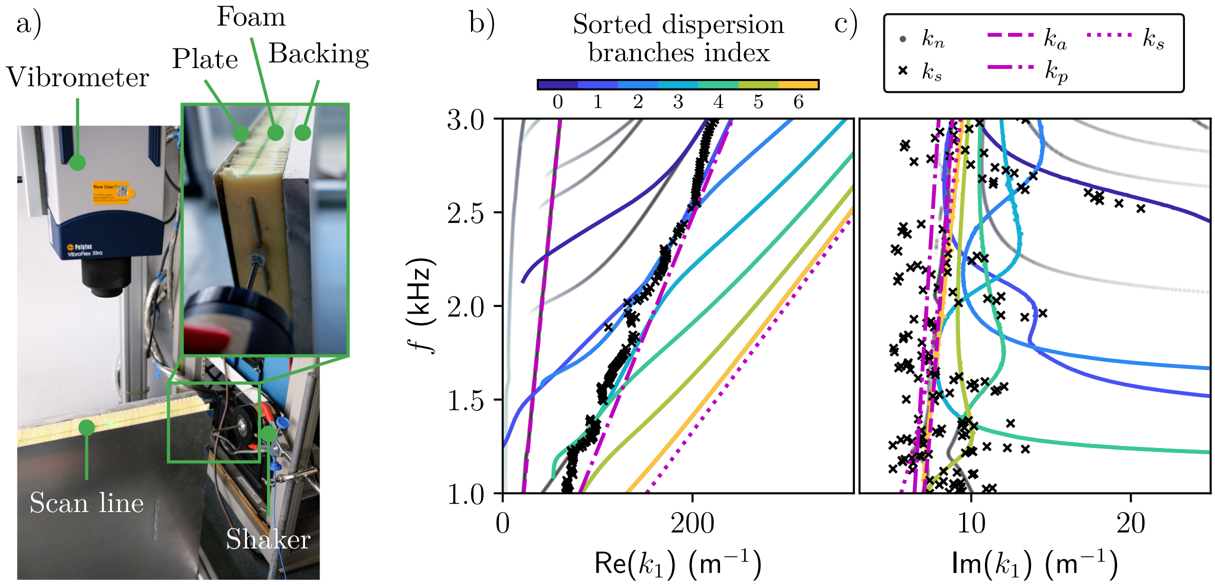

Fig. 3.4: a) Experimental set-up used for the measurement of the normal displacement field of the two-layer structure, with a close-up of the sample excitation. b) Real and c) imaginary parts of the dispersion relations recovered experimentally from the SLaTCoW method k_s (black crosses) and calculated from the SCM k_s (colored lines and black-to-grey dots) and the bulk wavenumbers in the poroelastic material (magenta lines).

numerical dispersion relation are not retrieved with the experimental results. This is because they correspond to a high attenuation of the mode, a part that is inherently difficult to measure in this experimental conditions. Generally, the recovered branches are those around the bulk solid compression wavenumber, corresponding to the wave that we most probably excite in the structure. The difficulties in efficiently exciting all the guided waves supported by this configuration largely explain the missing curves in the dispersion relation. The location and orientation of the excitation line limit possible symmetries and the generation of potential shear waves. This choice was made to avoid the main excitation of guided waves in the elastic plate when exitation and displacement measurement are performed only on this element. In addition, the modes mainly supported by the fluid phase should also be better excited by a loudspeaker and therefore better measured by a microphone. However, we have chosen to focus on the modes mainly supported by the solid phase. Nevertheless, the agreement between experimental results and simulation is quite good, thus validating the method.

3.5 Conclusion

A spectral collocation method (SCM) is proposed to calculate dispersion relations for guided acoustic wave propagation in multilayer structures. The formalism is general enough to be able to consider arbitrary arrangements of (visco-) elastic, poroelastic, and (viscothermal-) fluid layers. Outgoing fluid radiation, leading to Leaky modes, can be accounted for. The fields in each layer is approximated by Chebyshev polynomials and discretized on the corresponding nodes. Interface conditions are implemented to couple layers together. The solution provides the full complex wavenumber-real frequency dispersion relation of the considered structure. In addition, the physical fields in each layer are directly evaluated since they are the eigenvectors associated with the eigenvalue problem giving the dispersion relation. The dispersion diagram calculated with the present SCM is validated against experimental results and that calculated with a root-finding method (Müller method) in a simple two-layer structure consisting of a rigidly backed poroelastic layer covered by a thin aluminum plate radiating in a fluid half-space. The semi-analytical approach underlying the root-finding method is taken a step further to solve for the wave amplitudes as well, providing a reference for the mode shapes calculated with the SCM. The dispersion properties in such structures, as well as the direction of the energy density in each phase of the poroelastic layers using a decoupled expression for the Poynting vectors is analyzed. Results on energy transport velocity are also reported.

This method paves the way of complex wavenumber-real frequency dispersion relation calculation and analysis of more complex structures, like layers with embedded periodic inclusions (elastic or resonant), i.e. metaporolastic surfaces.

3.6 Appendices

3.6.A. Expression of the Biot elastic coefficients and effective parameters in poroelastic materials

The complex and frequency dependent density and bulk modulus of the fluid phase, which accounts respectively for the viscous and thermal losses are[105,106] as,

$$\begin{aligned}
\rho_{\mathrm{eq}}(\omega) &= \frac{\rho_f \alpha_\infty}{\phi}\left(1 + \frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty\omega}\sqrt{1 - \mathrm{i}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda}\right)^2}\right), \\
K_{\mathrm{eq}} &= \frac{\kappa P_0}{\phi}\left(\kappa - (\kappa-1)\Big/\left(1 + \frac{\mathrm{i}R_f\phi}{\rho_f\alpha_\infty \mathrm{Pr}\omega}\sqrt{1 - \mathrm{i}\mathrm{Pr}\omega\rho_f\mu_f\left(\frac{2\alpha_\infty}{R_f\phi\Lambda'}\right)^2}\right)\right)^{-1},
\end{aligned} \tag{3.28}$$

where ϕ is the porosity, α_∞ is the tortuosity, Λ and Λ' are respectively the viscous and thermal characteristic lengths, R_f is the flow resistivity, μ_f is the dynamic viscosity, κ is the heat capacity ratio, ρ_f the density of the saturating fluid, and P_0 is the ambient pressure.

The Biot elastic coefficients, P, Q, and R, are more commonly expressed in terms of the frame, K_b, effective fluid, K_{eq}, and shear, N_s, moduli when saturated by a light fluid as

$$P = K_b + \frac{4}{3}N_s + (1-\phi)^2\frac{K_{\mathrm{eq}}}{\phi}; \quad Q = K_f(1-\phi), \quad R = \phi K_{\mathrm{eq}}, \tag{3.29}$$

while the apparent densities are

$$\rho_{22} = \phi^2\rho_{eq}, \quad \rho_{12} = \phi\rho_f - \rho_{22}, \quad \rho_{11} = (1-\phi)\rho_s - \rho_{12}, \tag{3.30}$$

with ρ_s the density of the solid phase. The coefficients in Eq. (3.13) are thus

$$\hat{A} = K_b - \frac{2}{3}N_s + \frac{K_{\mathrm{eq}}}{\phi}\left((1 + (1-\phi)\right), \quad \hat{\rho} = \rho_1 + \phi\rho_f - 2\phi^2\rho_{eq} - \frac{\rho_f^2}{\rho_{eq}} \quad \beta = \frac{\rho_f}{\rho_{eq}} - 2\phi - 1, \tag{3.31}$$

where $\rho_1 = \phi\rho_f + (1-\phi)\rho_s$ is the apparent density of the frame. Finally, the frame bulk modulus and shear modulus are linked by the Poisson ratio ν with $K_b = 2N_s(1+\nu)/3(1-2\nu)$.

3.6.B. Application to a single poroelastic layer surrounded by two fluid half-spaces

The dispersion relation of guided waves in a single layer of poroelastic material surrounded by two identical fluid half-spaces is considered (See Fig.3.5a). The poroelastic material properties are

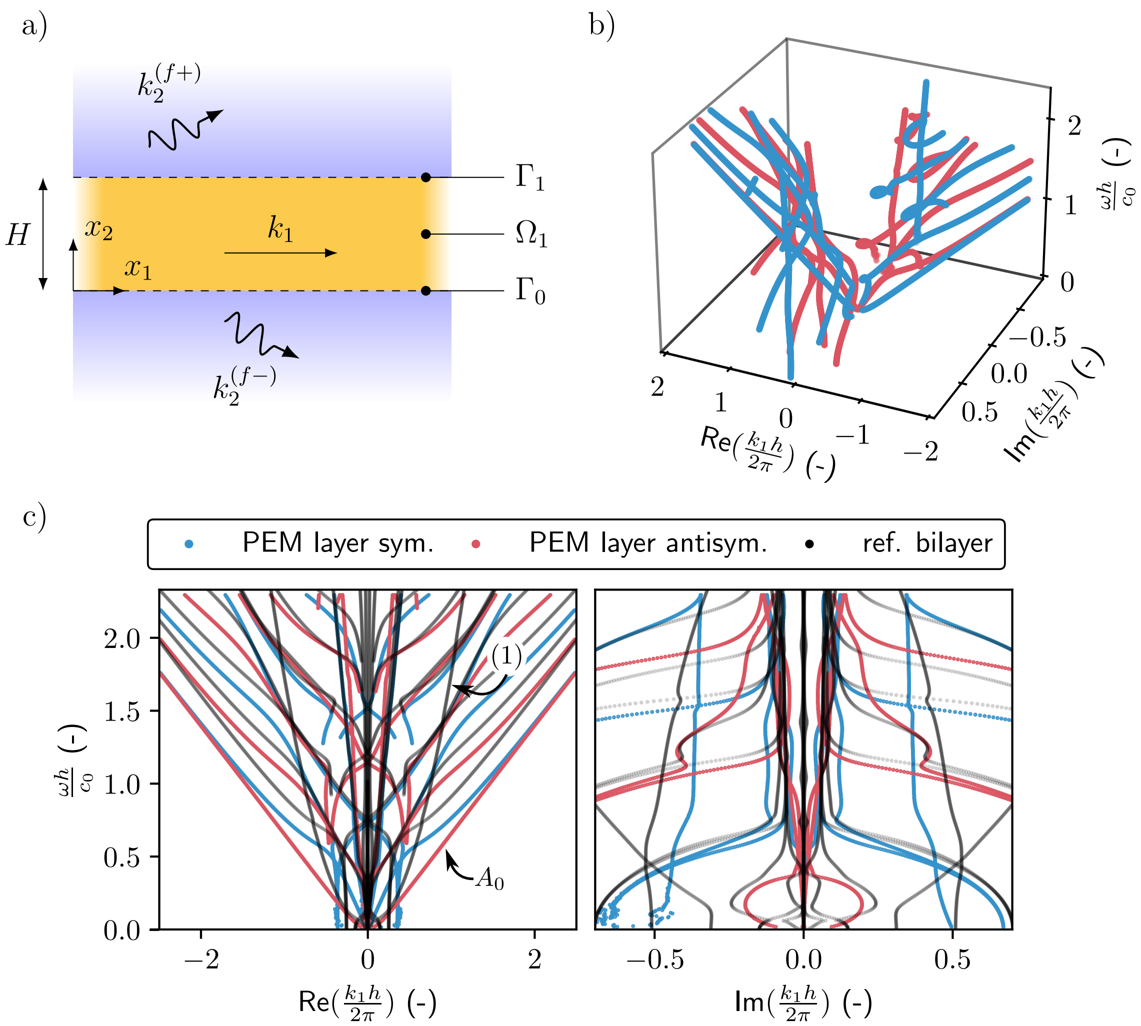

Fig. 3.5: a) Sketch of the single layer poroelastic structure geometry. b) 3D view of the complex wavenumber-real frequency dispersion relation and c) real and imaginary parts of the dispersion relation. The branches of the symmetric and antisymmetric modes are represented by blue and red curves respectively. The dispersion relation of the two-layer elatsic-poroelastic system studied in Section 3.4 are reminded with black lines.

those considered in Section 3.4.2 and the layer thickness is $2h^{(1)} = 104$ mm, i.e. twice the thickness of the poroelastic layer considered in the two-layer system.

The discretized equations of motion is that provided in Eq.(3.16). $M^{(1)} = 11$ collocation points are considered. Only the boundary conditions at the lower, i.e. Γ_0 at $x_2 = 0$, and upper, i.e. Γ_1 at $x_2 = h$, interfaces are modified. They read as[124]

$$\begin{aligned}
&\left(1+\phi+\phi\frac{\rho_{12}}{\rho_{22}}\right)u_2^{(1)} + \frac{\phi^2}{\rho_{22}\omega^2}\partial_2 p^{(1)} = u_2^{(0\pm)}, \qquad p^{(1)} = p^{(0\pm)}, \\
&\hat{\sigma}_{22}^s = \left(1-\phi\left(1+\frac{Q}{R}\right)\right)p^{(0\pm)}, \qquad \hat{\sigma}_{12}^s = 0.
\end{aligned} \tag{3.32}$$

The discretized form of these boundary conditions is

$$\underline{\underline{C}}_1 = \left(\begin{array}{ccc|c}
\frac{\phi^2}{\rho_{22}}\underline{\underline{D}}_2^+ & \underline{\underline{0}} & \omega^2\left(1+\phi\left(1+\frac{\rho_{12}}{\rho_{22}}\right)\right)\underline{\underline{I}}^+ & \frac{ik_2^{(f)}}{\rho_f} \\
\underline{\underline{I}}^+ & \underline{\underline{0}} & \underline{\underline{0}} & -1 \\
\underline{\underline{0}} & ik_1(\hat{P}-2N_s)\underline{\underline{I}}^+ & \hat{P}\underline{\underline{D}}_2^+ & 1-\phi\left(1+\frac{Q}{R}\right) \\
\underline{\underline{0}} & N_s\underline{\underline{D}}_2^+ & ik_1N_s\underline{\underline{I}}^+ & 0
\end{array}\right). \tag{3.33}$$

These matrices are then arranged in the same manner as Eq.(3.10) and the wavenumbers are calculated. Fig. 3.5b-c depicts the results.

In a similar way as Lamb modes, the different modes can be sorted in symmetric and antisymmetric modes. This is done by comparing the sign of the radiated amplitudes at each edge of the layer. Results are shown in Fig. 3.5b. All modes necessarily have an imaginary part, both because of the radiation condition, i.e. leakage, and because of viscothermal dissipation as can be noticed in Fig. 3.5c.

3.6.C. Evaluation of the physical fields from the SCM solutions

The discrete sets of eigenvectors resulting from the SCM are the spatial Laplace transforms of some physical fields at collocation points, i.e. the locations where the discrete equations of motion (residue) are exactly satisfied. These discrete fields are interpolated into a continuous form using

$$\boldsymbol{s}^{(n)}(x_2) = \left[\underline{\underline{\psi}}_m^{-1}(\underline{\xi})\underline{s}_m^{(n)}(\underline{\xi})\right]\psi\left(\xi^{(n)}\right) \tag{3.34}$$

where the terms in brackets correspond to the coefficients $\alpha_m^{(n)}$ given in Eq. (3.8). The Chebyshev polynomials ψ is the basis that maps from the discrete grid $\underline{\xi}$ back to the physical space.

The stress tensor components depend on the direct expressions of the fields and their first order derivative in x_2, denoted with a prime symbol in the following. The various fields, differentiated up to any order, can be calculated by replacing ψ by its derivative in Eq. (3.34). Explicit expressions are given in the following,

- in the elastic layer, the spatial Laplace transforms of stress tensor components are computed from $\boldsymbol{u}$, with

$$\begin{gathered}\sigma_{11} = (\lambda+2\mu)\mathrm{i}k_1\boldsymbol{u}_1 + \lambda\boldsymbol{u}_2', \quad \sigma_{12} = \mu\left(\boldsymbol{u}_1' + \mathrm{i}k_1\boldsymbol{u}_2\right), \\ \sigma_{22} = \lambda\mathrm{i}k_1\boldsymbol{u}_1 + (\lambda+2\mu)\boldsymbol{u}_2'.\end{gathered} \tag{3.35}$$

- in the poroelastic layer, the full solid stress tensor is expressed as $\boldsymbol{\sigma}_s = \hat{\boldsymbol{\sigma}}_s - \phi\left(Q/R\right)\boldsymbol{p}$, with $\hat{\boldsymbol{\sigma}}_s$ the spatial Laplace transform of the *in vacuo* solid stress tensor introduced in Section 3.4. The components of this full solid stress tensor are,

$$\begin{gathered}\sigma_{s,11} = P\mathrm{i}k_1\boldsymbol{u}_{s,1} + \hat{A}\boldsymbol{u}_{s,2}' - \phi\frac{Q}{R}\boldsymbol{p}, \quad \sigma_{s,12} = N_s\left(\boldsymbol{u}_{s,1}' + \mathrm{i}k_1\boldsymbol{u}_{s,2}\right), \\ \sigma_{s,22} = \hat{A}\mathrm{i}k_1\boldsymbol{u}_{s,2} + P\boldsymbol{u}_{s,2}' - \phi\frac{Q}{R}\boldsymbol{p}.\end{gathered} \tag{3.36}$$

  The Laplace transform of the fluid stress tensor is calculated as

$$\boldsymbol{\sigma}_f = -\phi\boldsymbol{p}\delta_{ij}, \tag{3.37}$$

  and the fluid displacement components are

$$\boldsymbol{u}_{f,1} = \frac{\phi}{\rho_{22}\omega^2}\mathrm{i}k_1\boldsymbol{p} - \frac{\rho_{12}}{\rho_{22}}\boldsymbol{u}_{s,1}, \quad \boldsymbol{u}_{f,2} = \frac{\phi}{\rho_{22}\omega^2}\boldsymbol{p}' - \frac{\rho_{12}}{\rho_{22}}\boldsymbol{u}_{s,2}. \tag{3.38}$$

All physical fields $\theta(\boldsymbol{x})$ can finally be calculated by the inverse spatial Laplace transform with $\theta(\boldsymbol{x}) = \tilde{\theta}(k_1, x_2)e^{\mathrm{i}k_1x_1}$.

3.6.D. Energy flux in the multilayer structure - Poynting theorem in a poroelastic medium

Energy conservation is commonly conducted in the time domain. Let us consider a volum Ω, bounded by $\partial\Omega$, with $\boldsymbol{n}$ the outgoing normal vector. The energy balance or Poynting theorem in a poroelastic materials reads as[82,131]

$$
\begin{aligned}
-\int_{\partial\Omega}(-\boldsymbol{\sigma}_s\cdot\boldsymbol{v}_s^\star)\cdot\boldsymbol{n}+(-\boldsymbol{\sigma}_f\cdot\boldsymbol{v}_f^\star)\cdot\boldsymbol{n}\,dS=\frac{1}{2}\int_\Omega &\rho_1\partial_t|\boldsymbol{v}_s|^2-\rho_{12}'\partial_t\left(\boldsymbol{v}_f-\boldsymbol{v}_s\right)^2+\rho_2\partial_t|\boldsymbol{v}_f|^2\,d\Omega\\
&+\int_\Omega \boldsymbol{\sigma}_s:\partial_t\boldsymbol{\epsilon}_s^\star+\boldsymbol{\sigma}_f:\partial_t\boldsymbol{\epsilon}_f^\star-b\left(\boldsymbol{v}_s-\boldsymbol{v}_f\right)^2\,d\Omega,
\end{aligned}
\tag{3.39}
$$

where $\boldsymbol{v}_s=\partial_t\boldsymbol{u}_s$ and $\boldsymbol{v}_f=\partial_t\boldsymbol{u}_f$ are the velocity fields, $\boldsymbol{\sigma}_s$ and $\boldsymbol{\sigma}_f$ are the stress tensors, $\boldsymbol{\epsilon}_s=1/2\left(\boldsymbol{\nabla}\boldsymbol{u}_s+\boldsymbol{\nabla}^\mathsf{T}\boldsymbol{u}_s\right)$ and $\boldsymbol{\epsilon}_f=1/2\left(\boldsymbol{\nabla}\boldsymbol{u}_f+\boldsymbol{\nabla}^\mathsf{T}\boldsymbol{u}_f\right)$ are the strain tensors in the solid frame and in the fluid phase respectively, and $\star$ denotes the complex conjugate. Note that a similar expression is provided in[129], where the alternative 1962 Biot formulation is used.

The left-hand side contains the energy density flux $\boldsymbol{P}_s=-\boldsymbol{\sigma}_s\cdot\boldsymbol{v}_s^\star$ and $\boldsymbol{P}_f=-\boldsymbol{\sigma}_f\cdot\boldsymbol{v}_f^\star$ in the elastic frame and fluid phase. The right-hand side contains the kinetic energy K[11] and the internal forces $P_{\text{int}}=\partial_t W+P_{dis}$. The latter is the sum of the time derivative of the strain energy W and the dissipated power P_{dis} by the elastic damping and the viscous and thermal losses[131]. In the frequency domain, the kinetic and strain energies read as

$$
\begin{aligned}
K&=\frac{1}{2}\rho_1\partial_t|\boldsymbol{v}_s|^2-\frac{1}{2}\rho_{12}'\partial_t\left(\boldsymbol{v}_f-\boldsymbol{v}_s\right)^2+\frac{1}{2}\rho_2\partial_t|\boldsymbol{v}_f|^2,\\
W&=\frac{1}{2}\hat{\boldsymbol{\sigma}}_s:\boldsymbol{\epsilon}_s^\star+\frac{1}{2}R\nabla\cdot(Q\boldsymbol{u}_s+R\boldsymbol{u}_f),
\end{aligned}
\tag{3.40}
$$

Finally, the time average total Poynting vector is $\langle\boldsymbol{P}_t\rangle=\langle\boldsymbol{P}_s\rangle+\langle\boldsymbol{P}_f\rangle$, with $\langle\boldsymbol{P}_s\rangle=-\mathrm{Re}\left(\boldsymbol{\sigma}_s\cdot\boldsymbol{v}_s^\star\right)/2$ and $\langle\boldsymbol{P}_f\rangle=-\mathrm{Re}\left(\boldsymbol{\sigma}_f\cdot\boldsymbol{v}_f^\star\right)/2$ and the total energy density is $\langle U\rangle=\langle K\rangle+\langle W\rangle=\mathrm{Re}(K)/2+\mathrm{Re}(W)/2$. Note that a similar procedure can be followed for isotropic elastic or fluid medium[132]. In particular, the elastic Poynting vector is $\langle\boldsymbol{P}\rangle=\mathrm{Re}\left(\boldsymbol{\sigma}\cdot\boldsymbol{v}^\star\right)/2$, where $\boldsymbol{\sigma}$ is the stress tensor and $\boldsymbol{v}$ is the velocity in the elastic medium.

The energy transport velocity V_e is evaluated from the latter expressions

$$
V_e=\frac{\langle\boldsymbol{P}_t\rangle\cdot\boldsymbol{n}}{\langle U\rangle}.
\tag{3.41}
$$

When a multi-layer configuration is considered, the latter velocity is averaged over the structure thickness $\overline{V_e}=\dfrac{\overline{P_t}}{\overline{U}}$, with the average Poynting vector and the average total energy defined as

$$
\overline{P}_t=\sum_n\frac{1}{h_n}\int_{h_{n-1}}^{h_n}\langle\boldsymbol{P}_t^{(n)}\rangle\cdot\boldsymbol{e}_{x_2}\,dx_2,\quad \overline{U}=\sum_n\frac{1}{h_n}\int_{h_{n-1}}^{h_n}\langle U^{(n)}\rangle\,dx_2.
\tag{3.42}
$$

3.6.E. Reminder of the SLaTCoW method for obtaining complex-valued wavenumbers.

The key point of the SLaTCoW method is the Laplace transform, $\mathfrak{L}_{\text{mes}}(z)$ with $z=z_r+\mathrm{i}z_i$ the Laplace space coordinate, of the data measured in space at a fixed frequency ω. In doing so, we move from the spatial domain to the complex wavenumber domain. The difference between $\mathfrak{L}_{\text{mes}}(z)$

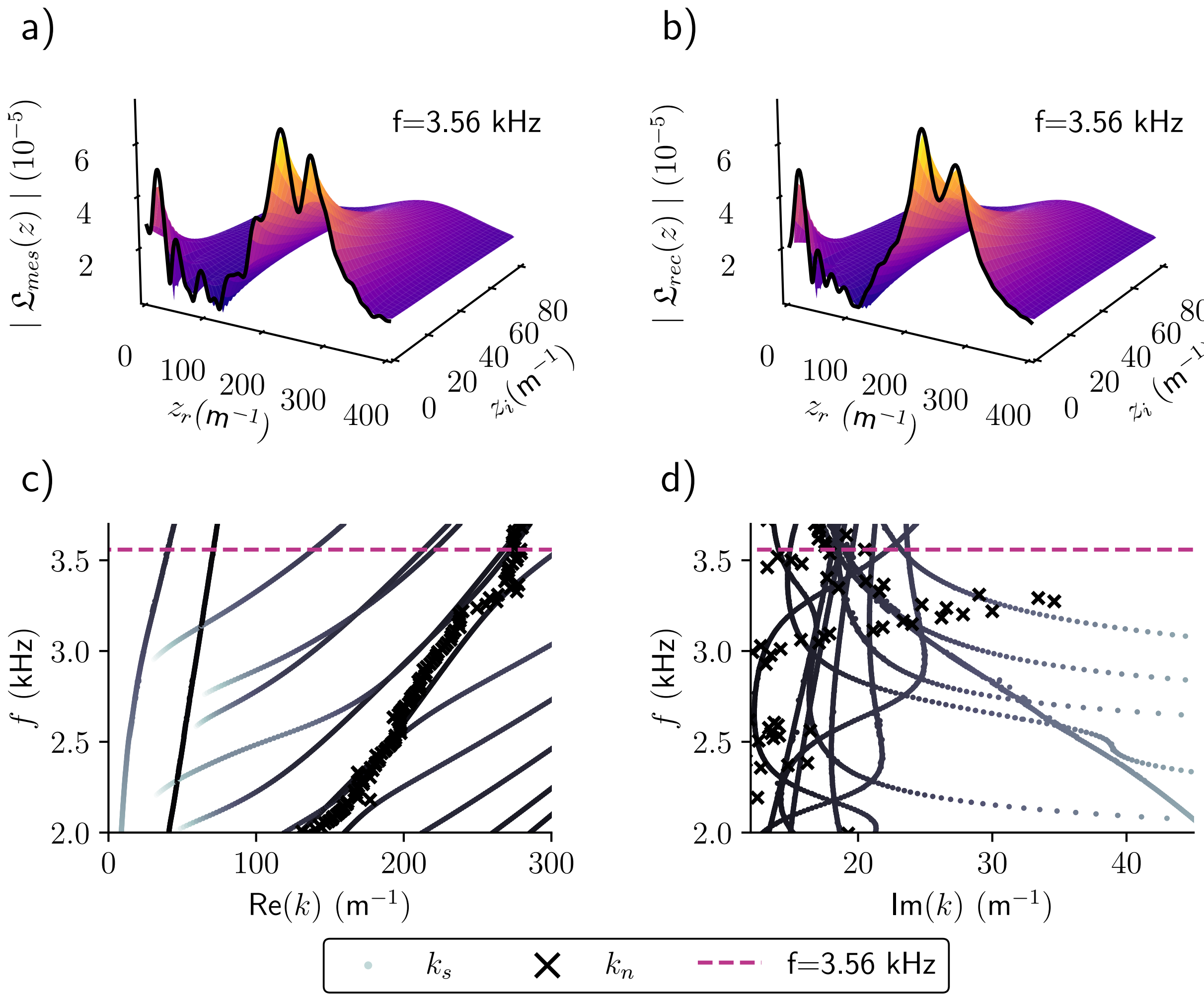

Fig. 3.6: Description of the procedure of the SLaTCoW method: a) Spatial Laplace transform from the experimental data, b) Spatial Laplace transform of the reconstructed wave fields. Real c) and imaginary d) parts of the experimentally retrieved wavenumbers (crosses) compared to the SCM (black-to-grey dots). The frequency at which the Laplace transforms are represented corresponds to the magenta dashed line in the graph.

and the spatial Laplace transform of the displacement ansatz along the same line, $\mathfrak{L}_{\text{rec}}(z)$, is then minimized in the least square sense to recover both the complex amplitude (magnitude and phase) and the complex wavenumber of each mode assumed to be part of the displacement ansatz.

The experimental Laplace spatial transform $\mathfrak{L}_{\text{mes}}(z) = \int_0^L r(x)e^{-\mathrm{i}zx}dx$ is computed at each frequency ω from the transfer function $r(x)$ between the excitation and the normal displacement measured along a line $x \in [0, L]$, with $L = 0.3$ m and spacing $d_x = 0.001$ m. The displacement anstaz is assumed to be of the form $w(x) = \sum_{\beta} A^{\beta} e^{\mathrm{i}k^{\beta}x}$, with β the number of guided-waves, i.e., the number of modes, and where $A^{\beta} = |A^{\beta}|e^{\mathrm{i}\phi^{\beta}}$ is the complex amplitude and k^{β} is the complex wavenumber of the β-th mode.. In practice, β is the smallest integer that gives a good fit between the two Laplace transforms and therefore depends on the frequency under consideration. Its spatial Laplace transform reads as $\mathfrak{L}_{\text{rec}}(z) = \sum_{\beta} A^{\beta} L e^{\mathrm{i}(k^{\beta}-z)\frac{L}{2}} \operatorname{sinc}\left((k^{\beta} - z)\frac{L}{2}\right)$.

The cost function $\mathfrak{C} = \left(\|\mathfrak{L}_{\text{rec}}(z) - \mathfrak{L}_{\text{mes}}(z)\|\right) / \|\mathfrak{L}_{\text{mes}}(z)\|$ is finally minimized using the Matlab function *fminsearchbnd* for $\{|A^{\beta}|,\ \phi^{\beta},\ \operatorname{Re}\left(k^{\beta}\right),\ \operatorname{Im}\left(k^{\beta}\right)\}, \forall \beta$. The initial guess are the recovered values of the minimization parameters at the previous frequency, with the exception of the first frequency. The method is graphically exposed in Figure 3.6. Figures 3.6 c) and d) depict the complex dispersion relation as already depicted in Fig. 3.4. Figure 3.6a) depicts the spatial Laplace transform of the experimental data and Figure 3.6b) depicts $\mathfrak{L}_{\text{rec}}(z)$ with the parameters derived from minimization of the cost function at the frequency $f = 3.56$ kHz. This frequency is highlighted by the purple dashed line in Figures 3.6 c) and d). The two Laplace transforms match almost perfectly.

Appendices

Contents

A Implementing SAFE to compute the dispersion relation of Lamb modes

As discussed in Section. 3.2, the SAFE method is an alternative to SCM for the numerical computation of dispersion relation of guided waves in layers. They rely on a similar formalism, that is a 1D discretizatin across the thickness of each layer and an assumption of guided harmonic waves along the longitudinal direction. An analogous method to that presented in Chapter 3 could be developed using SAFE, the key difference being the location of nodes in the layer and the interpolation function used for discretization. There is a large bibliography dealing with these topics and applying this method to similar problem to ours.

The general theory of SAFE is explained in great details in Chapter 9 of Rose's textbook[94]. The present method is inspired from this material. Examples of SAFE application to dispersion curve calculation are illustrated by Ref.[133] where a numerical model was implemented to investigate periodic structure of infinite width with the addition of periodic boundary conditions allowing for the computation of guided wave modes. A similar approach was generalized for systems with arbitrary cross-sections and a coupling with infinite solid media by using the so-called absorbing region[134]. Other description of the surrounding medium where leakage happens have also been investigated by using a PML in Ref.[135]. This has been done by Mazzotti *et al.* using BEM[136,137]. A local coupling of the surrounding space was done in Ref.[138] by linearizing a non-linear GEP, a similar approach to that of Refs.[P1,102].

In this appendix, we propose to quickly re-derive a SAFE method inspired from these previous papers to benchmark the two numerical methods regarding their respective performances. We will use high-order Lobatto function as interpolating function of the finite-elements core of the method. The system studied here is the simple of Lamb modes: an elastic layer of infinite extent in the x_1 direction has two free edges at $x_2 = 0$ and $x_2 = h$, where a no-traction boundary condition is applied.

A.1. Formalism

Let us start with the general variational form for the elasticity equation

$$-\rho\omega^2 \int_\Omega u\delta u \, d\Omega + \int_\Omega \boldsymbol{\sigma}(u) : \boldsymbol{\epsilon}(\delta u) \, d\Omega = \int_{\partial\Omega} \boldsymbol{\sigma}(\delta u) \cdot \boldsymbol{n} \, d\Gamma, \quad \forall \, \delta u, \tag{A.1}$$

where the displacement fields is

$$\boldsymbol{u}(\boldsymbol{x}) = \boldsymbol{u}(x_2) e^{ik_1 x_1} e^{-i\omega t}, \tag{A.2}$$

and the double-dot product is $\boldsymbol{\sigma}(u) : \boldsymbol{\epsilon}(\delta u^\star) = \sigma_{ij}(u)\varepsilon_{ij}(\delta u^\star)$. $\boldsymbol{\sigma}$ designates the stress tensor while $\boldsymbol{\epsilon}$ is the strain tensor. Developing this part of the volume term gives the expression,

$$\begin{aligned}
\sigma_{11}\varepsilon_{11} &= k_1^2(\lambda + 2\mu)u_1\delta u_1 + -ik_1\lambda\nabla u_2\delta u_1 \\
\sigma_{12}\varepsilon_{12} &= \frac{\mu}{2}\left(-ik_1\nabla u_1\delta u_2 + \nabla u_1\delta\nabla u_1 + k_1^2 u_2\delta u_2 + ik_1 u_2\nabla\delta u_1)\right) \\
\sigma_{22}\varepsilon_{22} &= ik_1\lambda u_1\nabla\delta u_2 + (\lambda + 2\mu)\nabla u_2\nabla\delta u_2.
\end{aligned} \tag{A.3}$$

Putting these terms back into Eq. A.1 yields,

$$\begin{aligned}
\int_\Omega & -\rho\omega^2\,(u_1\delta u_1 + u_2\delta u_2) + k_1^2(\lambda + 2\mu)u_1\delta u_1 - ik_1\lambda\nabla u_2\delta u_1 - ik_1\mu\nabla u_1\delta u_2 + \mu\nabla u_1\delta\nabla u_1 \\
& + \mu k_1^2 u_2\delta u_2 + ik_1\mu u_2\nabla\delta u_1) + ik_1\lambda u_1\nabla\delta u_2 + (\lambda + 2\mu)\nabla u_2\nabla\delta u_2 \, d\Omega = 0 \quad \forall \, \delta u
\end{aligned} \tag{A.4}$$

where ∇u_i denotes differentiation of the i-th component of the displacement along x_2. Note that the surface term initally expressed vanishes from the no-traction boundary condition.

The displacement fields can be discretized as

$$\boldsymbol{u}(\boldsymbol{x}) = \boldsymbol{N}(\xi)\boldsymbol{U} e^{ik_1 x_1} e^{-i\omega t}, \tag{A.5}$$

with $\boldsymbol{N}$ containing the Lobatto shape functions, and $\boldsymbol{U}$ is the vector of displacement amplitudes. Following the same discretization, the test function thus becomes $\forall\delta\boldsymbol{U}$. By rearranging the terms of Eq. A.4 and writing the weak form for one element,

$$\begin{aligned}
\int_{\Omega_e} & k_1^2\Big(\mu\delta\boldsymbol{U}_2^\top\boldsymbol{M}^{(e)}\boldsymbol{U}_2 + (\lambda + 2\mu)\delta\boldsymbol{U}_1^\top\boldsymbol{M}^{(e)}\boldsymbol{U}_1\Big) + ik_1\Big(\mu\delta\boldsymbol{U}_2^\top\boldsymbol{C}_1^{(e)}\boldsymbol{U} - \mu\delta\boldsymbol{U}_1^\top\boldsymbol{C}_2^{(e)}\boldsymbol{U}_2 \\
& + \lambda\delta\boldsymbol{U}_1^\top\boldsymbol{C}_1^{(e)}\boldsymbol{U}_2 - \lambda\delta\boldsymbol{U}_2^\top\boldsymbol{C}_2^{(e)}\boldsymbol{U}_1\Big) + \mu\delta\boldsymbol{U}_1^\top\boldsymbol{K}^{(e)}\boldsymbol{U}_1 + (\lambda + 2\mu)\delta\boldsymbol{U}_2^\top\boldsymbol{K}^{(e)}\boldsymbol{U}_2 \\
& - \rho\omega^2\delta\boldsymbol{U}_1^\top\boldsymbol{M}^{(e)}\boldsymbol{U}_1 - \rho\omega^2\delta\boldsymbol{U}_2^\top\boldsymbol{M}^{(e)}\boldsymbol{U}_2 \, d\Omega_e = 0
\end{aligned} \tag{A.6}$$

where the elementary matrices are,

$$\begin{aligned}
\boldsymbol{M}^{(e)} &= \frac{h}{2}\int_{-1}^{1} \boldsymbol{N}^\top\boldsymbol{N} \, d\xi, \quad & \boldsymbol{K}^{(e)} &= \frac{2}{h}\int_{-1}^{1} \nabla\boldsymbol{N}^\top\nabla\boldsymbol{N} d\xi, \\
\boldsymbol{C}_1^{(e)} &= \int_{-1}^{1} \boldsymbol{N}^\top\nabla\boldsymbol{N} \, d\xi, \quad & \boldsymbol{C}_2^{(e)} &= \int_{-1}^{1} \nabla^\top\boldsymbol{N}^\top\boldsymbol{N}\boldsymbol{U} \, d\xi,
\end{aligned} \tag{A.7}$$

In Eq. A.6, the dependance on k_1 was highlighted and terms were sorted depending on their order. The elementary matrices should be assembled into the global matrices and then, following an analogous procedure to Chapter. 3, the general form for the problem becomes,

$$\left(k_1^2\boldsymbol{A}_2 + ik_1\boldsymbol{A}_1 + \boldsymbol{A}_0 - \omega^2\boldsymbol{M}\right)\boldsymbol{U} = 0. \tag{A.8}$$

It can easily be rewritten as a generalized eigenvalue problem,

$$\left[\begin{pmatrix} -\boldsymbol{A}_1 & \boldsymbol{A}_0 - \omega^2 \boldsymbol{M} \\ \boldsymbol{I} & \boldsymbol{0} \end{pmatrix} - k_1 \begin{pmatrix} \boldsymbol{A}_2 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{pmatrix}\right] \begin{pmatrix} k_1 \boldsymbol{U} \\ \boldsymbol{U} \end{pmatrix} = 0, \tag{A.9}$$

which solving is thus quite straightforward.

A.2. Results and convergence comparison

Analytical results are obtained by applying the root-finding approach using the Müller method presented in the paper. Solutions are sought from the classical Lamb modes equations, derived in many books such as Royer. Figure 3.7a) shows the results of the three methods for a given frequency range. They all match well. In order to compare the two numerical methods in play, a brief convergence study has been done using the analytical results from the root-finding method as a reference.

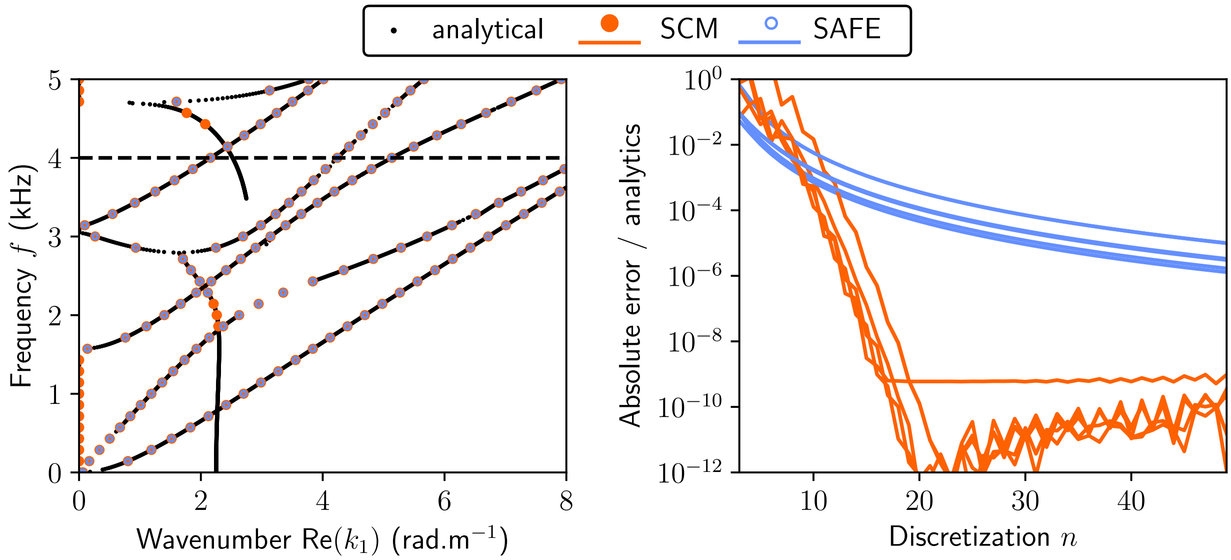

Fig. 3.7: a) Dispersion relation for the Lamb modes with the root-finding results, alongside the spectral collocation and semi-analytical finite elements solutions. b) Convergence curve of the 2 methods at a given frequency $f = 4$ kHz. The multiple lines depicts the convergence of each wavenumber.

The convergence curve of the results given in Fig. 3.7b) is unsurprisingly in agreement with the literature. It is calculated at a frequency $f = 4$ kHz: the solutions along the dashed line in Fig 3.7a) are selected. The parameter n used for the discretization is shared between the two methods and corresponds to the number of elements taken for the discretization in each method. SAFE matrices have a size $2(2n + 1)$, with while SCM matrices have a size $2n$. The discretization is swept from $n = 3$ to $n = 50$.

The SCM convergence shows a steep slope and reaches a plateau around $n = 20$, while the SAFE curve follows the expected slope of a finite elements method that use second-order shape functions. When comparing at a given discretization order, the error from the SAFE is much lower that that of SAFE by several order of magnitudes. In order to see comparable convergence, it would require to use higher-order shape functions or use spectral elements instead.

B Details on the Müller method

B.1. The root-finding algorithm

equations, iterative scheme etc.
139

$$
\begin{aligned}
\Sigma_x &= \prod_{0\le i<j\le 2} (x_j - x_i) = (x_0 - x_2)(x_1 - x_2)(x_0 - x_1) \\
a &= \frac{(x_1 - x_2)(f_0 - f_2) - (x_0 - x_2)(f_1 - f_2)}{\Sigma_x} \\
b &= \frac{(x_0 - x_2)^2(f_1 - f_2) - (x_1 - x_2)^2(f_0 - f_2)}{\Sigma_x} \\
c &= f_2
\end{aligned}
\tag{B.1}
$$

$$
f(x_3) = a x_3^2 + b x_3 + c \tag{B.2}
$$

$$
x_3 = x_2 - \frac{2c}{b + \mathrm{sign}(b)\sqrt{b^2 - 4ac}} \tag{B.3}
$$

$$
\varepsilon_a = \left| \frac{x_3 - x_2}{x_3} \right| \tag{B.4}
$$

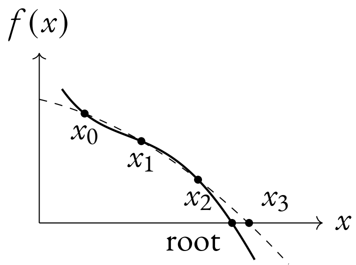

Fig. 3.8: iteration muller

B.2. A complete routine for the automated solving of dispersion relations

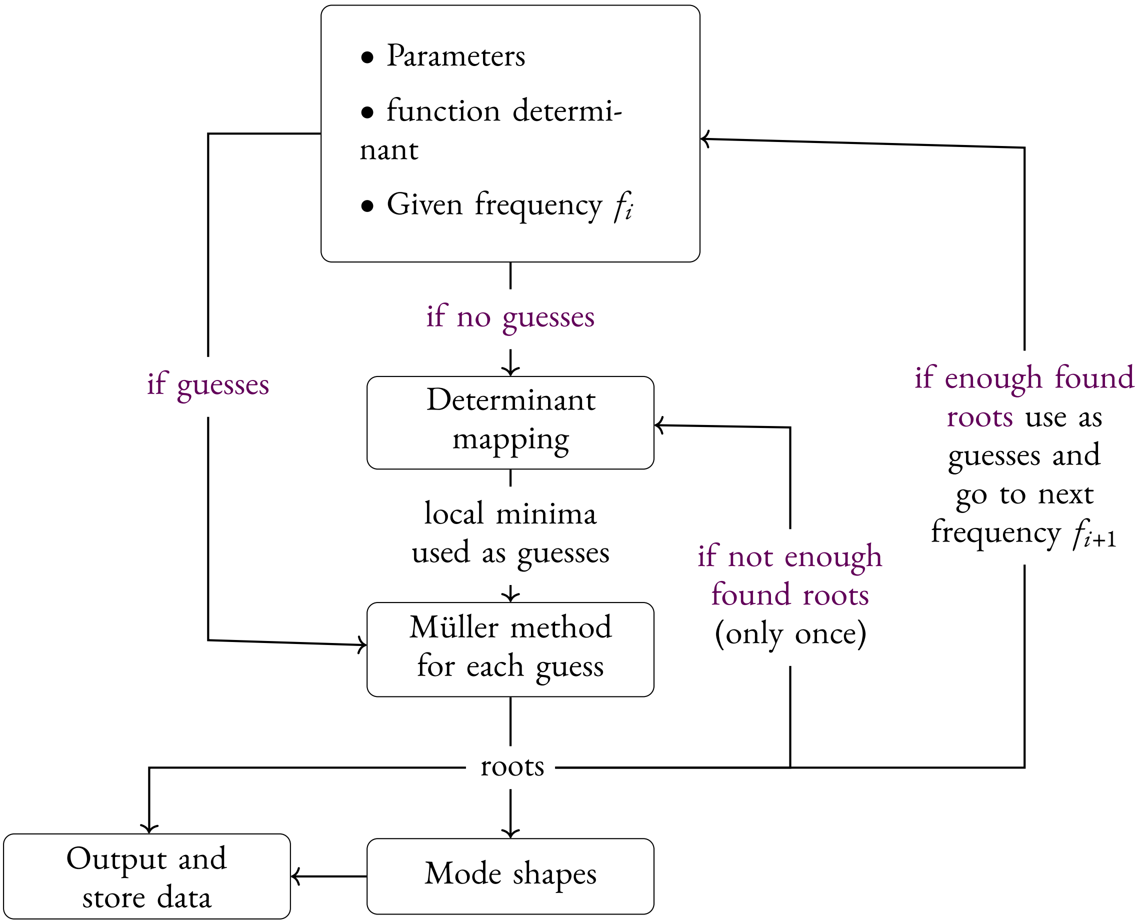

Fig. 3.9: Description of the routine

C | Overview of activities and scientific works during the thesis

C.1. Publications

P1. M. Maréchal et al. "A General Spectral Collocation Method for Computing the Dispersion Relations of Guided Acoustic Waves in Multilayer Dissipative Structures". *J. App. Phys.* 137.10 (2025), 104902.

C.2. Conferences

(ajouter titres des présentations)

— SAM 2022 Nimes, France, October 2022 (attendee)

— JJCAB Toulouse 2023, France, July 2023 (poster)

— SAPEM Sorrento, Italy, October 2023 (presenter)

— Séminaire de 2e année Centrale Nantes, France, April 2024 (presenter)

— SAM Aegina, Greece, May 2024 (presenter)

— INTERNOISE Nantes, France, August 2024 (attendee)

— CFA 2025 Paris, France (presenter)

C.3. Workshops and trainng schools

- training school Acoustic Metamaterials Valencia Spain, UPV, November 2023
- lamb waves workshop

C.4. Teaching activities

D Extended abstract in french

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